Physics for Scientists and Engineers Storyline

Tenth Edition

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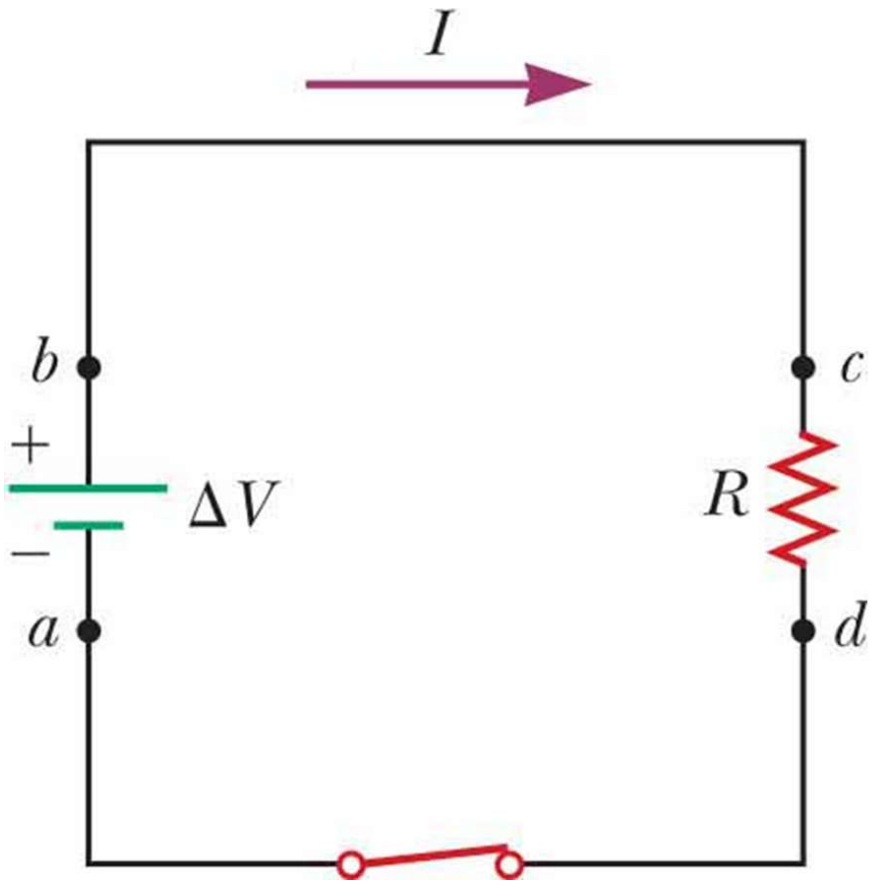
Chapter 27

Direct Current Circuits

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Electromotive Force (1 of 2)



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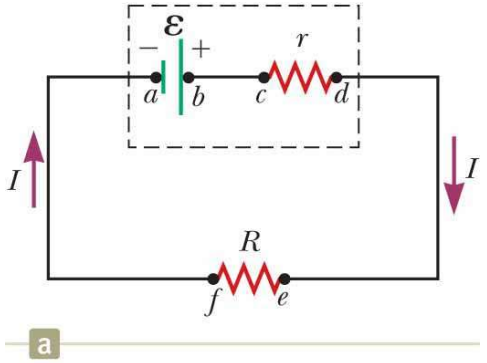
Electromotive Force (2 of 2)

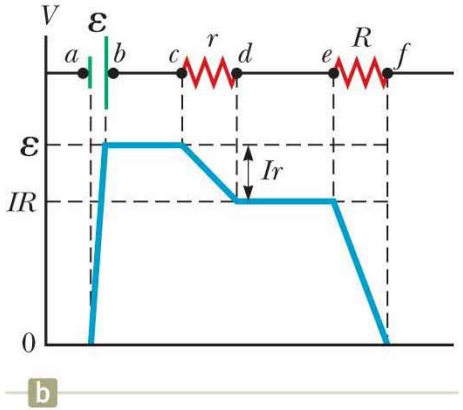
$$\Delta V = \varepsilon - Ir$$

$$\varepsilon = IR + Ir$$

$$I = \frac{\varepsilon}{R + r}$$

$$I\varepsilon = I^2 R + I^2 r$$





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Quick Quiz 27.1 (1 of 2)

To maximize the percentage of the power from the emf of a battery that is delivered to a device external to the battery, what should the internal resistance of the battery be?

(a) It should be as low as possible.

(b) It should be as high as possible.

(c) The percentage does not depend on the internal resistance.

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Quick Quiz 27.1 (2 of 2)

To maximize the percentage of the power from the emf of a battery that is delivered to a device external to the battery, what should the internal resistance of the battery be?

(a) It should be as low as possible.

(b) It should be as high as possible.

(c) The percentage does not depend on the internal resistance.

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Example 27.1: Terminal Voltage of a Battery (1 of 3)

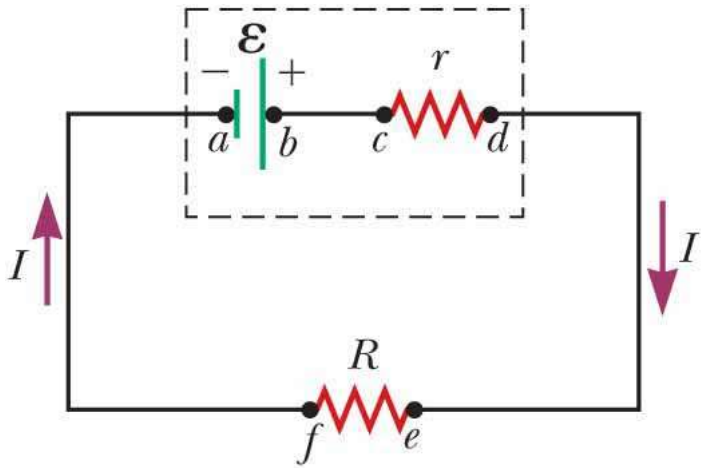
A battery has an emf of 12.0 V and an internal resistance of 0.050 0 W. Its terminals are connected to a load resistance of 3.00 Ω.

(A) Find the current in the circuit and the terminal voltage of the battery.

$$I = \frac{\varepsilon}{R+r} = \frac{12.0\ \text{V}}{3.00\ \Omega + 0.050\,0\ \Omega} = 3.93\ \text{A}$$

$$\begin{aligned}\Delta V &= \varepsilon - Ir \\ &= 12.0\ \text{V} - (3.93\ \text{A})(0.050\,0\ \Omega) \\ &= \boxed{11.8\ \text{V}}\end{aligned}$$

$$\Delta V = IR = (3.93\ \text{A})(3.00\ \Omega) = 11.8\ \text{V}$$



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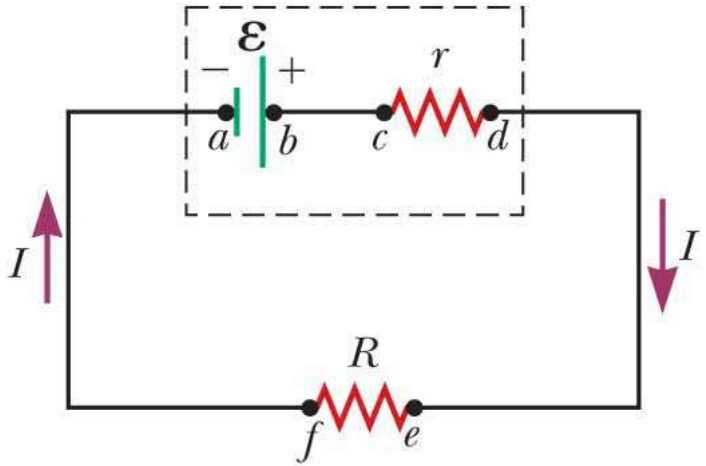
Example 27.1: Terminal Voltage of a Battery (2 of 3)

(B) Calculate the power delivered to the load resistor, the power delivered to the internal resistance of the battery, and the power delivered by the battery.

$$P_R = I^2 R = (3.93\ \text{A})^2 (3.00\ \Omega) = \boxed{46.3\ \text{W}}$$

$$P_r = I^2 r = (3.93\ \text{A})^2 (0.0500\ \Omega)$$
$$= \boxed{0.772\ \text{W}}$$

$$P = P_R + P_r = 46.3\ \text{W} + 0.772\ \text{W}$$
$$= \boxed{47.1\ \text{W}}$$



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Example 27.1: Terminal Voltage of a Battery (3 of 3)

As a battery ages, its internal resistance increases. Suppose the internal resistance of this battery rises to 2.00 Ω toward the end of its useful life. How does that alter the battery's ability to deliver energy?

$$I = \frac{\varepsilon}{R + r} = \frac{12.0\ \text{V}}{3.00\ \Omega + 2.00\ \Omega} = 2.40\ \text{A}$$

$$\Delta V = \varepsilon - Ir = 12.0\ \text{V} - \left(2.40\ \text{A}\right)\left(2.00\ \Omega\right) = 7.2\ \text{V}$$

$$P_R = I^2 R = \left(2.40\ \text{A}\right)^2 \left(3.00\ \Omega\right) = 17.3\ \text{W}$$

$$P_r = I^2 r = \left(2.40\ \text{A}\right)^2 \left(2.00\ \Omega\right) = 11.5\ \text{W}$$

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Example 27.2: Matching the Load (1 of 2)

Find the load resistance *R* for which the maximum power is delivered to the load resistance in the figure.

$$P = I^2 R = \frac{\varepsilon^2 R}{(R+r)^2}$$

$$\frac{dP}{dR} = \frac{d}{dR}\left[\frac{\varepsilon^2 R}{(R+r)^2}\right] = \frac{d}{dR}\left[\varepsilon^2 R(R+r)^{-2}\right] = 0$$

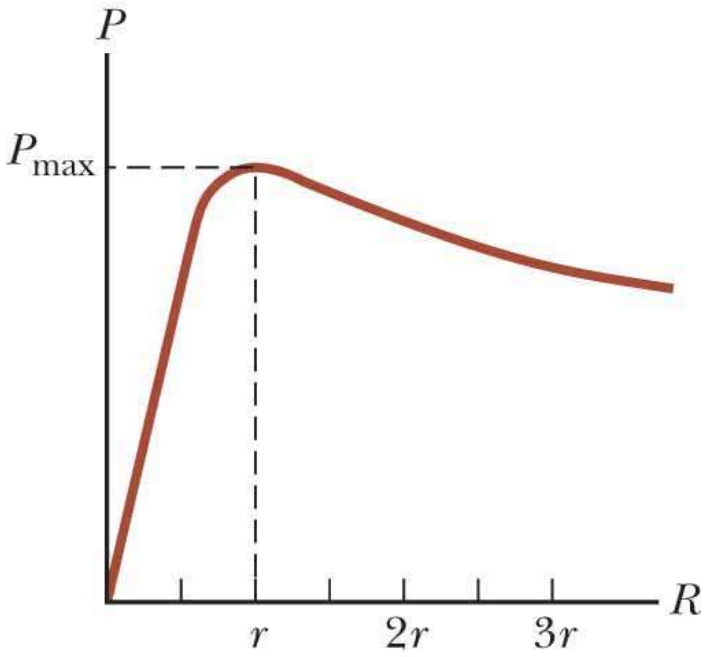
$$\left[\varepsilon^2 R(R+r)^{-2}\right] + \left[\varepsilon^2 R(-2)(R+r)^{-3}\right] = 0$$

$$\frac{\varepsilon^2(R+r)}{(R+r)^3} - \frac{2\varepsilon^2 R}{(R+r)^3} = \frac{\varepsilon^2(r-R)}{(R+r)^3} = 0 \qquad \Rightarrow R = \boxed{r}$$

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Example 27.2: Matching the Load (2 of 2)



$$P = I^2 R = \frac{\varepsilon^2 R}{(R+r)^2}$$

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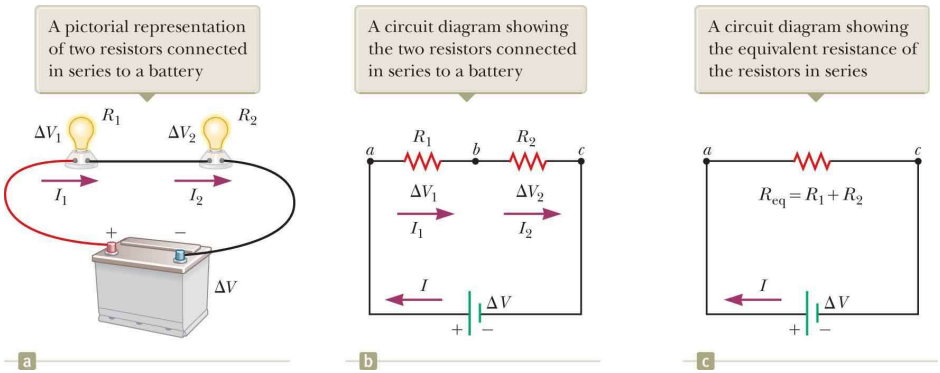
Resistors in Series and Equivalent Resistance

$$I = I_1 = I_2$$

$$\Delta V = \Delta V_1 + \Delta V_2 = I_1 R_1 + I_2 R_2$$

$$\Delta V = I R_{\text{eq}}$$

$$\boxed{R_{\text{eq}} = R_1 + R_2 + R_3 + \cdots}$$



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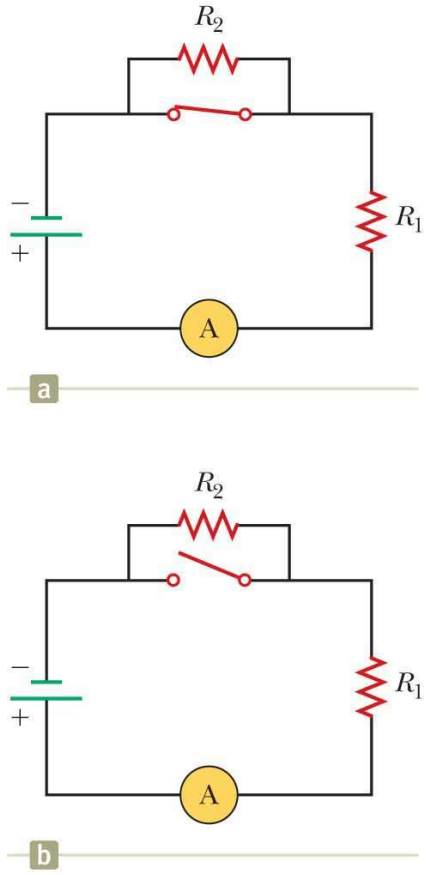
Quick Quiz 27.2 (1 of 2)

With the switch in the circuit of the top figure closed, there is no current in R_2 because the current has an alternate zero-resistance path through the switch. There is current in R_1, and this current is measured with the ammeter (a device for measuring current) at the bottom of the circuit. If the switch is opened (the bottom figure), there is current in R_2. What happens to the reading on the ammeter when the switch is opened?

(a) The reading goes up.

(b) The reading goes down.

(c) The reading does not change.



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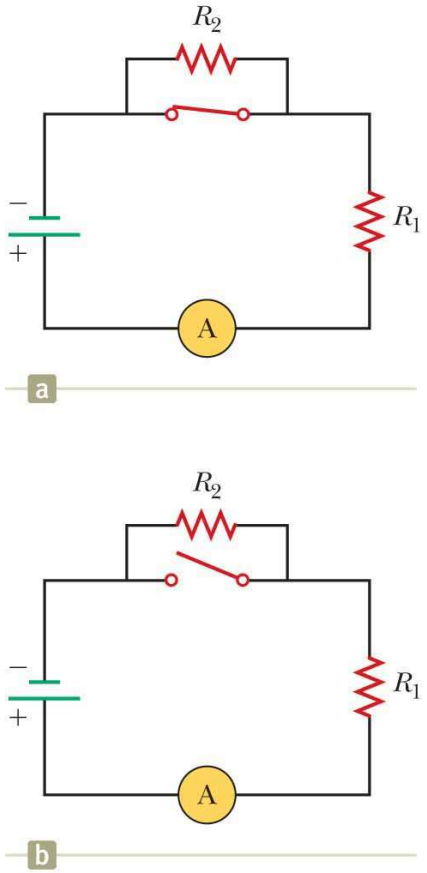
Quick Quiz 27.2 (2 of 2)

With the switch in the circuit of the top figure closed, there is no current in R_2 because the current has an alternate zero-resistance path through the switch. There is current in R_1, and this current is measured with the ammeter (a device for measuring current) at the bottom of the circuit. If the switch is opened (the bottom figure), there is current in R_2. What happens to the reading on the ammeter when the switch is opened?

(a) The reading goes up.

(b) The reading goes down.

(c) The reading does not change.



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Resistors in Parallel

$$\Delta V = \Delta V_1 = \Delta V_2$$

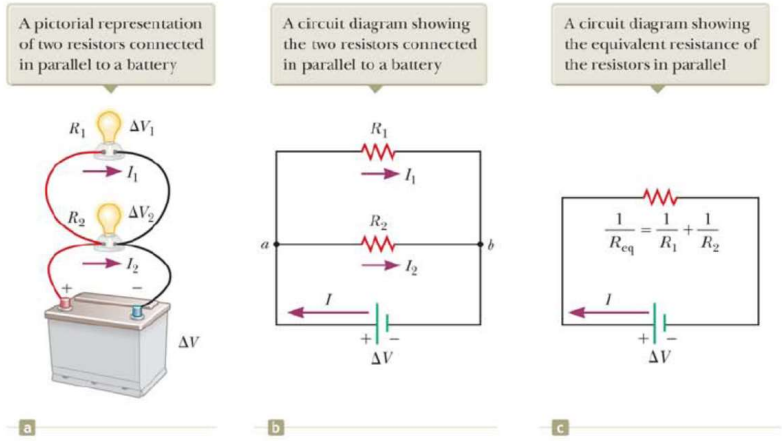
$$I = I_1 + I_2 = \frac{\Delta V_1}{R_1} + \frac{\Delta V_2}{R_2}$$

$$I = \frac{\Delta V}{R_{\text{eq}}}$$

$$\frac{\Delta V}{R_{\text{eq}}} = \frac{\Delta V_1}{R_1} + \frac{\Delta V_2}{R_2} \Rightarrow$$

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\boxed{\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots}$$



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Opening Storyline Revisited



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Cows, Chickens, and Lightning





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Birds on a Power Wire



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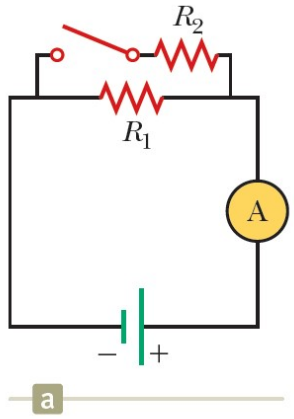
Quick Quiz 27.3 (1 of 2)

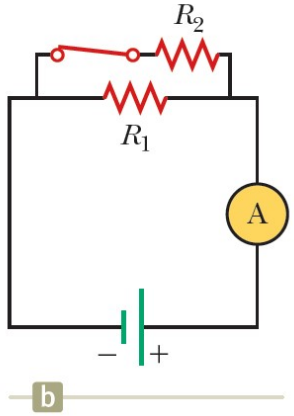
With the switch in the circuit of the top figure is open, there is no current in R_2. There is current in R_1, however, and it is measured with the ammeter at the right side of the circuit. If the switch is closed (the bottom figure), there is current in R_2. What happens to the reading on the ammeter when the switch is closed?

(a) The reading increases.

(b) The reading decreases.

(c) The reading does not change.





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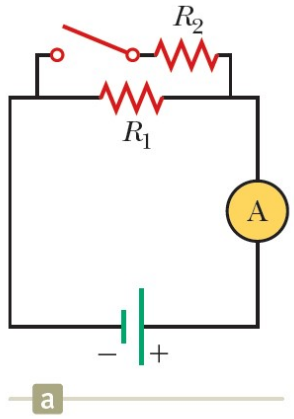
Quick Quiz 27.3 (2 of 2)

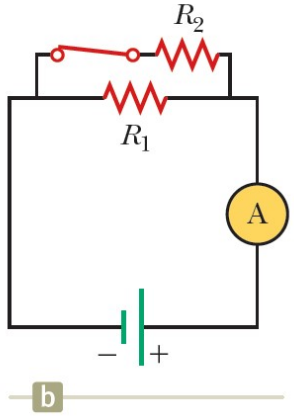
With the switch in the circuit of the top figure is open, there is no current in R_2. There is current in R_1, however, and it is measured with the ammeter at the right side of the circuit. If the switch is closed (the bottom figure), there is current in R_2. What happens to the reading on the ammeter when the switch is closed?

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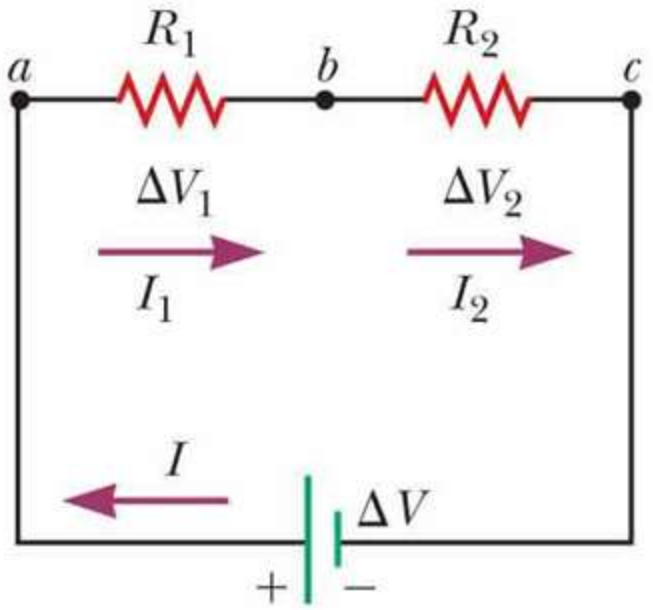
Quick Quiz 27.4 Part I (1 of 2)

In the figure, a third resistor is added in series with the first two. What happens to the current in the battery?

(a) increases

(b) decreases

(c) remains the same



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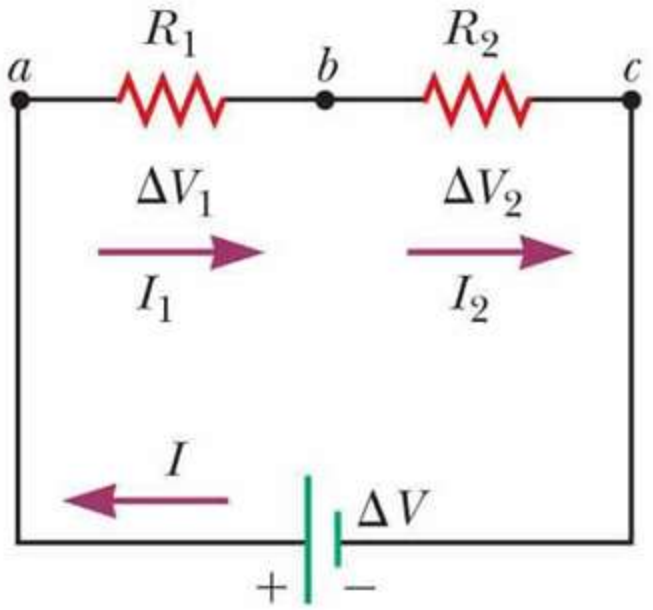
Quick Quiz 27.4 Part I (2 of 2)

In the figure, a third resistor is added in series with the first two. What happens to the current in the battery?

(a) increases

(b) decreases

(c) remains the same



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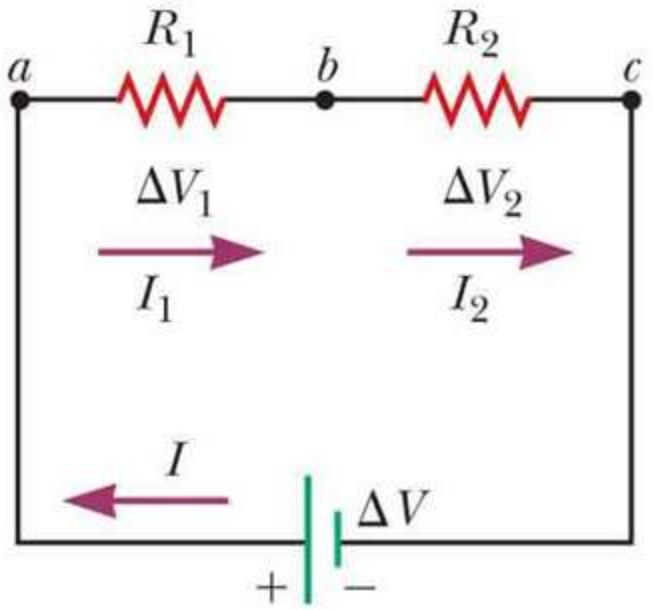
Quick Quiz 27.4 Part II (1 of 2)

What happens to the terminal voltage of the battery?

(a) increases

(b) decreases

(c) remains the same



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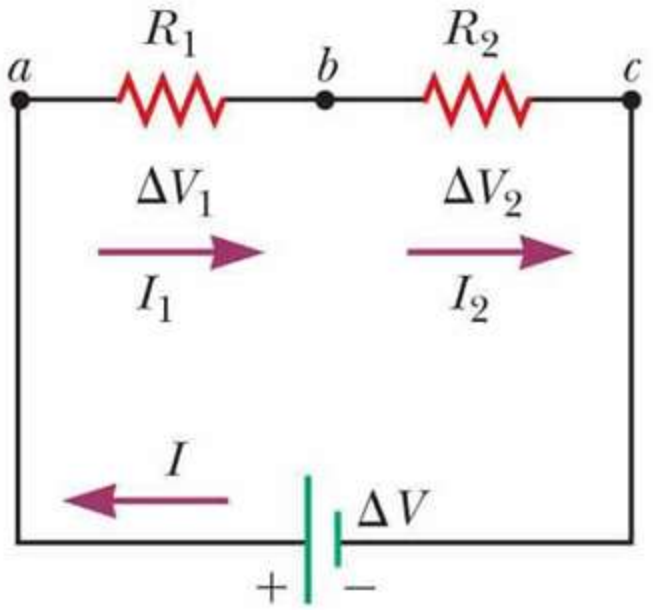
Quick Quiz 27.4 Part II (2 of 2)

What happens to the terminal voltage of the battery?

(a) increases

(b) decreases

(c) remains the same



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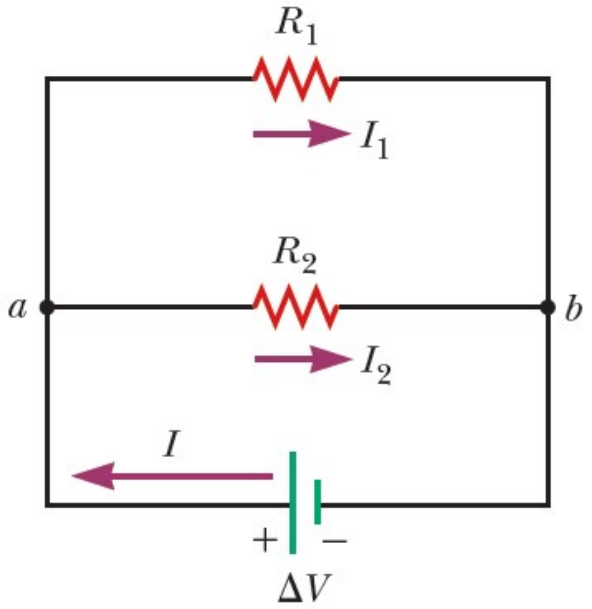
Quick Quiz 27.4 Part III (1 of 2)

In the figure, a third resistor is added in parallel with the first two. What happens to the current in the battery?

(a) increases

(b) decreases

(c) remains the same



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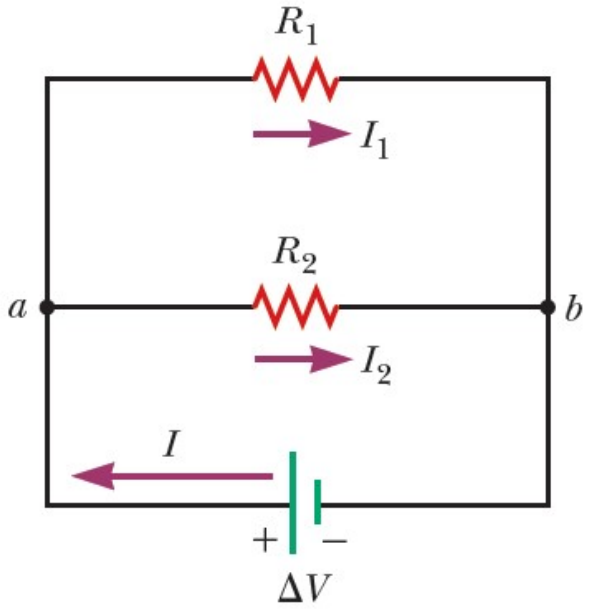
Quick Quiz 27.4 Part III (2 of 2)

In the figure, a third resistor is added in parallel with the first two. What happens to the current in the battery?

(a) increases

(b) decreases

(c) remains the same



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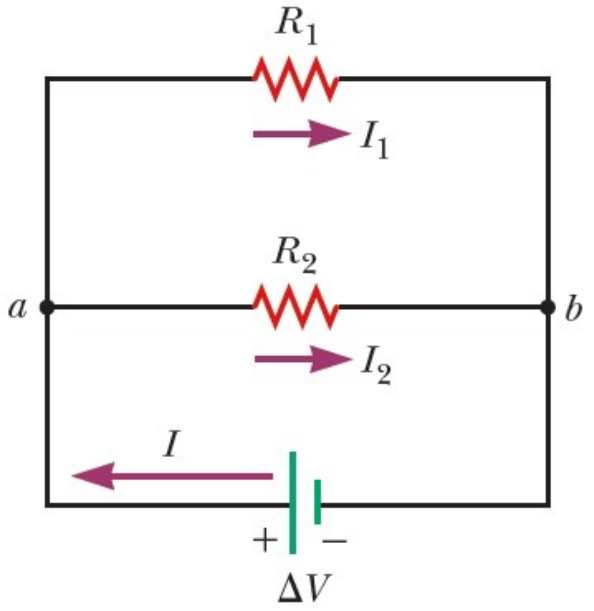
Quick Quiz 27.4 Part IV (1 of 2)

What happens to the terminal voltage of the battery?

(a) increases

(b) decreases

(c) remains the same



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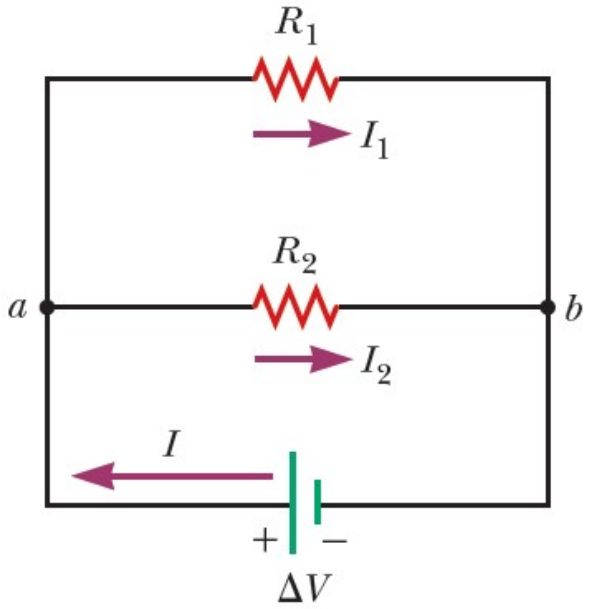
Quick Quiz 27.4 Part IV (2 of 2)

What happens to the terminal voltage of the battery?

(a) increases

(b) decreases

(c) remains the same



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Example 27.3: Landscape Lights (1 of 2)

A homeowner wishes to install low-voltage landscape lighting in his back yard. To save money, he purchases inexpensive 18-gauge cable, which has a relatively high resistance per unit length. This cable consists of two side-by-side wires separated by insulation, like the cord on an appliance. He runs a 200-foot length of this cable from the power supply to the farthest point at which he plans to position a light fixture. He attaches light fixtures across the two wires on the cable at 10-foot intervals so that the light fixtures are in parallel. Because of the cable's resistance, the brightness of the lightbulbs in the fixtures is not as desired. Which of the following problems does the homeowner have?

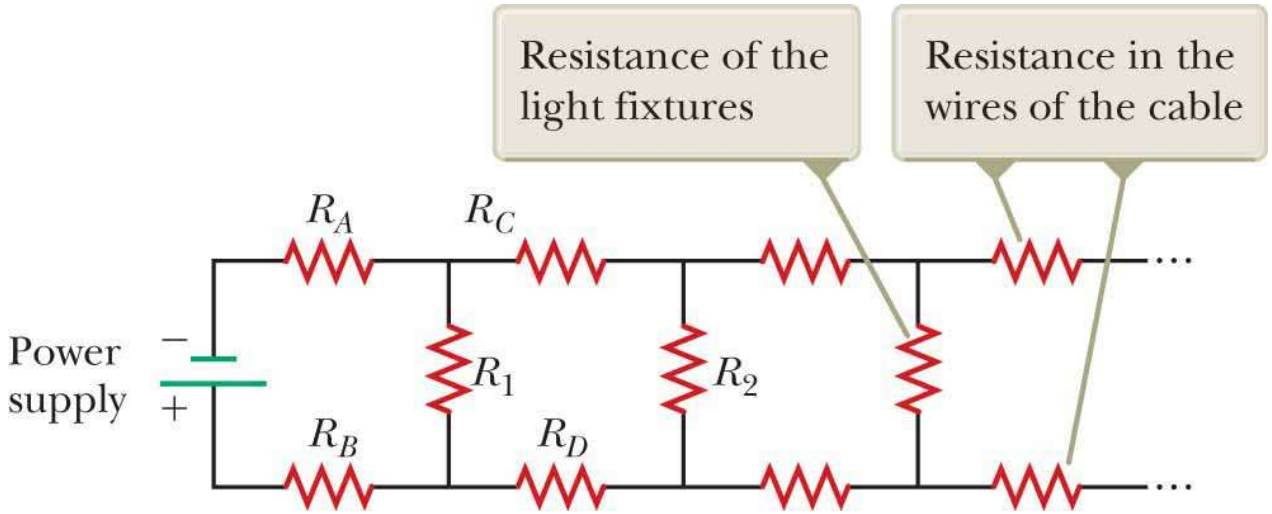
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Example 27.3: Landscape Lights (2 of 2)

(a) All the lightbulbs glow equally less brightly than they would if lower-resistance cable had been used.

(b) The brightness of the lightbulbs decreases as you move farther from the power supply.



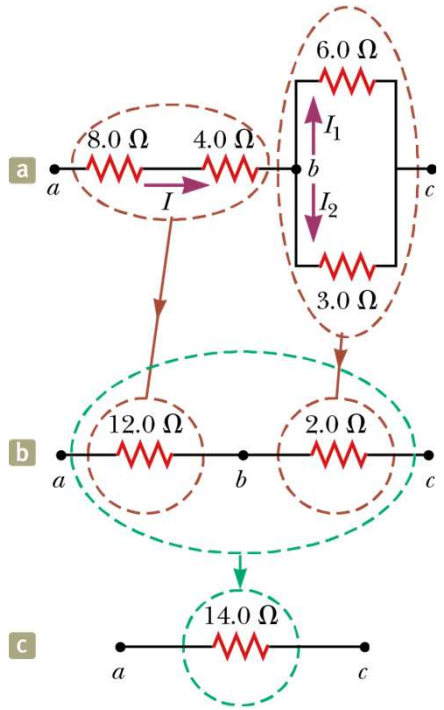
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Example 27.4: Find the Equivalent Resistance (1 of 3)

Four resistors are connected as shown in the figure.

(A) Find the equivalent resistance between points *a* and *c*.



$$R_{eq} = 8.0\ \Omega + 4.0\ \Omega = 12.0\ \Omega$$

$$\frac{1}{R_{eq}} = \frac{1}{6.0\ \Omega} + \frac{1}{3.0\ \Omega} = \frac{3}{6.0\ \Omega}$$

$$R_{eq} = \frac{6.0\ \Omega}{3} = 2.0\ \Omega$$

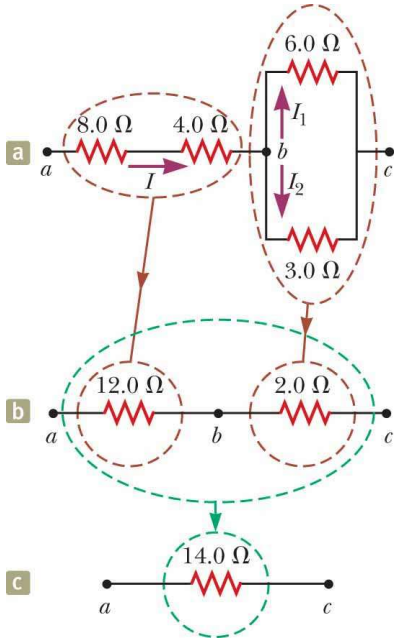
$$R_{eq} = 12.0\ \Omega + 2.0\ \Omega$$

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Example 27.4: Find the Equivalent Resistance (2 of 3)

(B) What is the current in each resistor if a potential difference of 42 V is maintained between *a* and *c*?



$$I = \frac{\Delta V_{ac}}{R_{\text{eq}}} = \frac{42\ \text{V}}{14.0\ \Omega} = \boxed{3.0\ \text{A}}$$

$$\Delta V_1 = \Delta V_2 \Rightarrow (6.0\ \Omega) I_1 = (3.0\ \Omega) I_2$$

$$\Rightarrow I_2 = 2I_1$$

$$I_1 + I_2 = 3.0\ \text{A} \Rightarrow I_1 + 2I_2 = 3.0\ \text{A} \Rightarrow I_1 = \boxed{1.0\ \text{A}}$$

$$I_2 = 2I_2 = 2(1.0\ \text{A}) = \boxed{2.0\ \text{A}}$$

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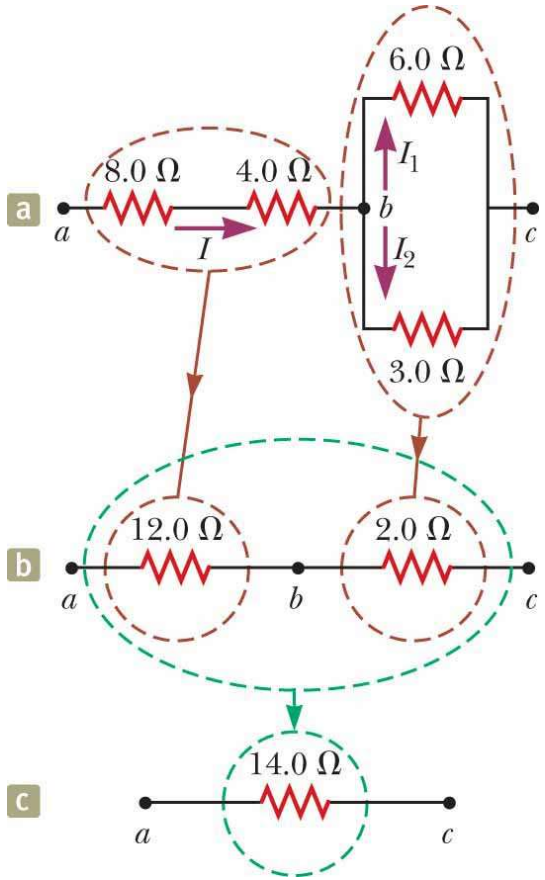
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Example 27.4: Find the Equivalent Resistance (3 of 3)

$$\Delta V_{bc} = (6.0\ \Omega) I_1 = (3.0\ \Omega) I_2 = 6.0\ \text{V}$$

$$\Delta V_{ab} = (12.0\ \Omega) I = 36\ \text{V}$$

$$\Delta V_{ac} = \Delta V_{Ab} + \Delta V_{bc} = 42\ \text{V}$$



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Example 27.5: Three Resistors in Parallel (1 of 4)

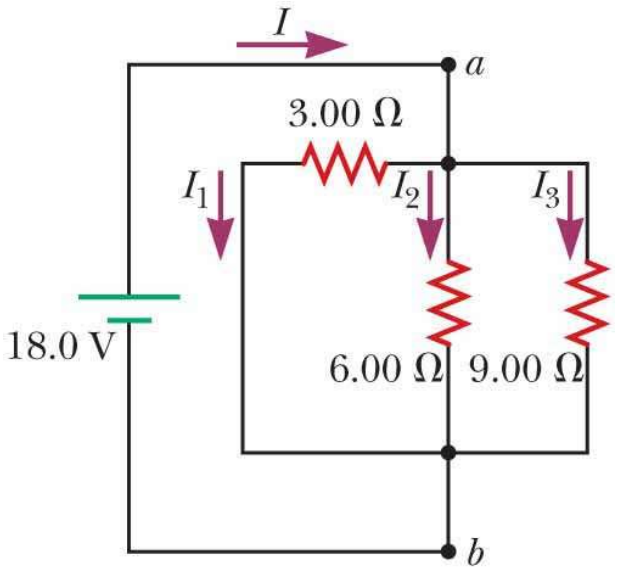
Three resistors are connected as shown in the figure. A potential difference of 18.0 V is maintained between points *a* and *b*.

(A) Calculate the equivalent resistance of the circuit.

$$\frac{1}{R_{eq}} = \frac{1}{3.00\ \Omega} + \frac{1}{6.00\ \Omega} + \frac{1}{9.00\ \Omega}$$

$$= \frac{11}{18.0\ \Omega}$$

$$\frac{1}{R_{eq}} = \frac{18.0\ \Omega}{11} = \boxed{1.64\ \Omega}$$



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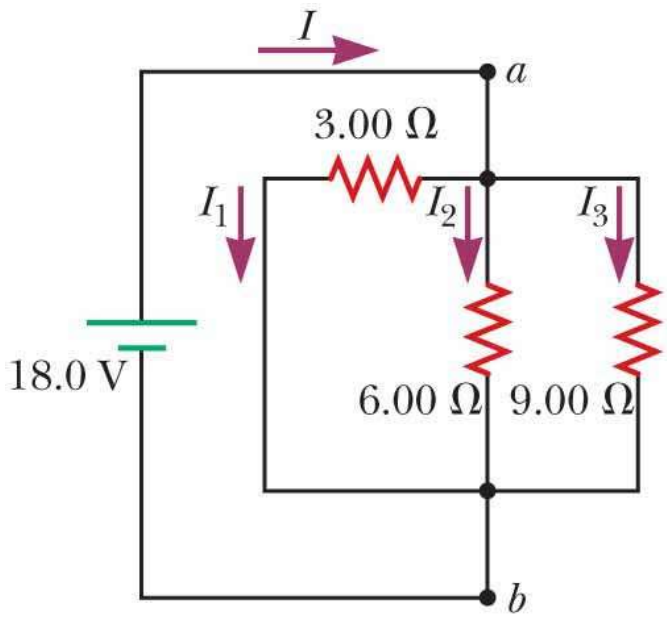
Example 27.5: Three Resistors in Parallel (2 of 4)

(B) Find the current in each resistor.

$$I_1 = \frac{\Delta V}{R_1} = \frac{18.0\ \text{V}}{3.00\ \Omega} = \boxed{6.00\ \text{A}}$$

$$I_2 = \frac{\Delta V}{R_1} = \frac{18.0\ \text{V}}{6.00\ \Omega} = \boxed{3.00\ \text{A}}$$

$$I_3 = \frac{\Delta V}{R_1} = \frac{18.0\ \text{V}}{9.00\ \Omega} = \boxed{2.00\ \text{A}}$$



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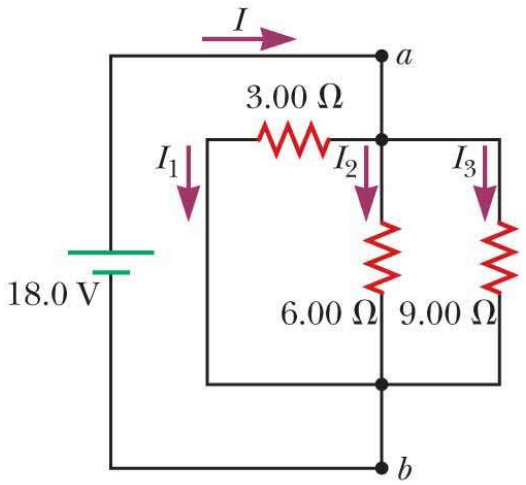
Example 27.5: Three Resistors in Parallel (3 of 4)

(C) Calculate the power delivered to each resistor and the total power delivered to the combination of resistors.

$$3.00\text{-}\Omega: P_1 = I_1^2 R_1$$
$$= (6.00\ \text{A})^2 (3.00\ \Omega) = \boxed{108\ \text{W}}$$
$$6.00\text{-}\Omega: P_2 = I_2^2 R_2$$
$$= (3.00\ \text{A})^2 (6.00\ \Omega) = \boxed{54\ \text{W}}$$
$$9.00\text{-}\Omega: P_3 = I_3^2 R_3$$
$$= (2.00\ \text{A})^2 (9.00\ \Omega) = \boxed{36\ \text{W}}$$

$$P = \frac{(\Delta V)^2}{R_{\text{eq}}} = \frac{(18.0\ \text{V})^2}{1.64\ \Omega} = 198\ \text{W}$$

$$P_T = (108 + 54 + 36)\ \text{W}$$
$$= 198\ \text{W}$$

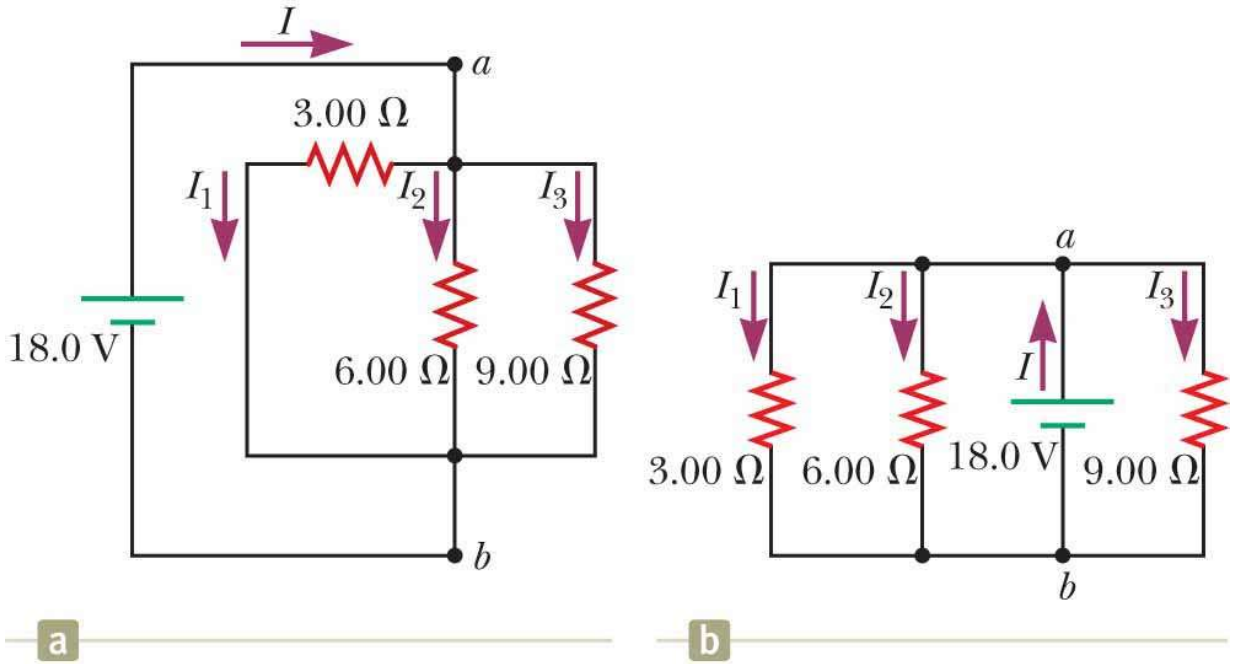


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Example 27.5: Three Resistors in Parallel (4 of 4)

What if the circuit were as shown in figure (b) instead? How would that affect the calculation?

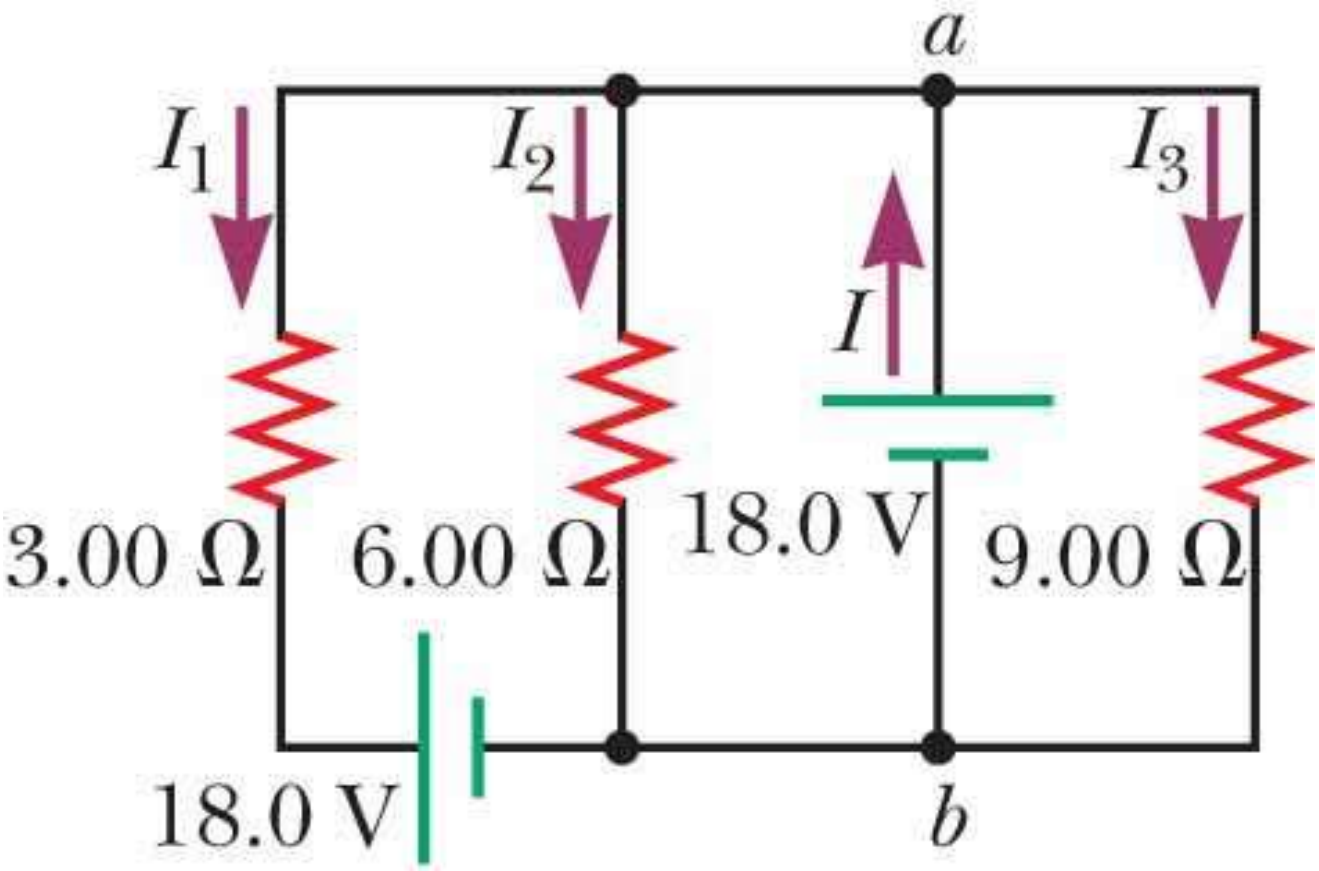


a b

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Kirchhoff's Rules (1 of 2)



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Kirchhoff's Rules (2 of 2)

1. Junction rule. At any junction, the sum of the currents must equal zero:

$$\sum_{\text{junction}} I = 0$$

2. Loop rule. The sum of the potential differences across all elements around any closed circuit loop must be zero:

$$\sum_{\text{closed loop}} \Delta V = 0$$

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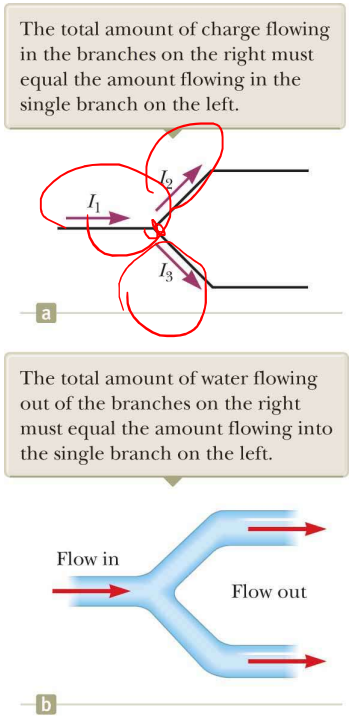
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Junction Rule

1. Junction rule. At any junction, the sum of the currents must equal zero:

$$\sum_{\text{junction}} I = 0$$

$$I_1 - I_2 - I_3 = 0$$



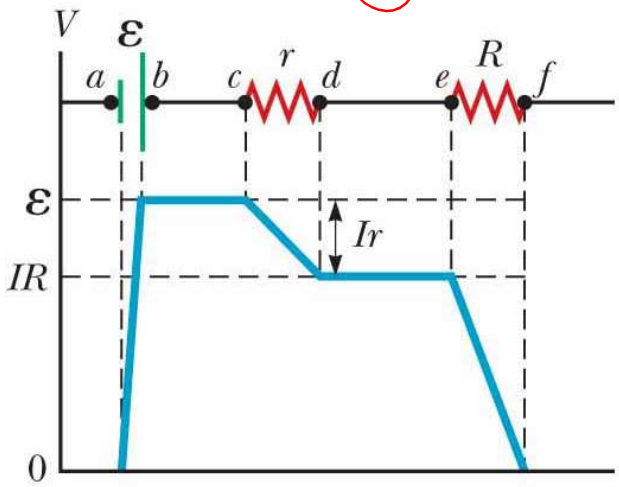
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Loop Rule

2. Loop rule. The sum of the potential differences across all elements around any closed circuit loop must be zero:

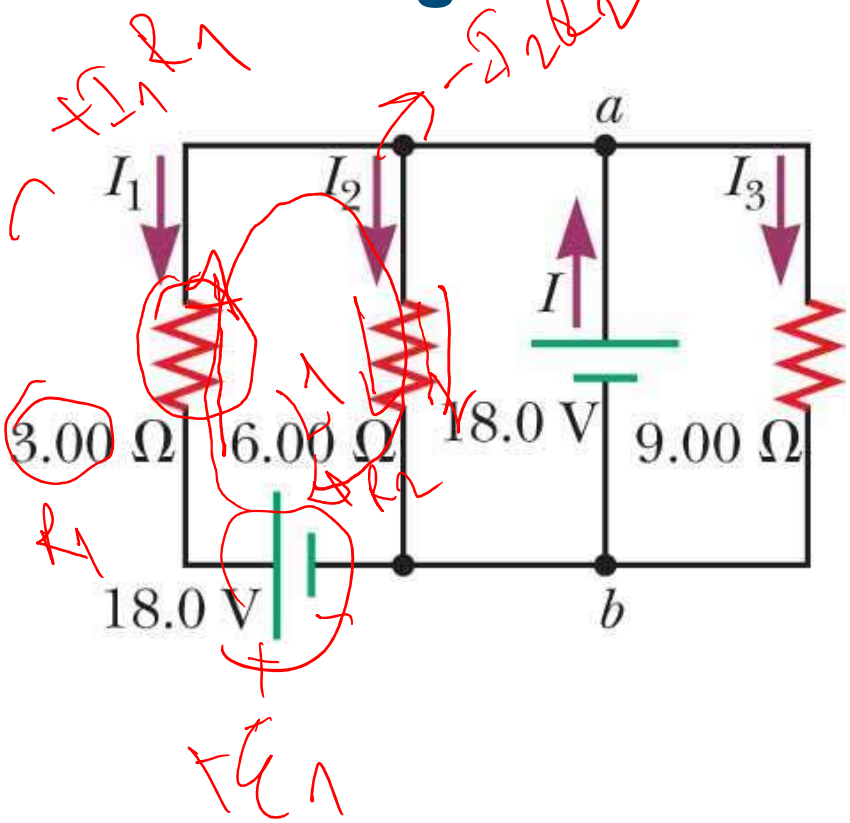
$$\sum_{\text{closed loop}} \Delta V = 0$$

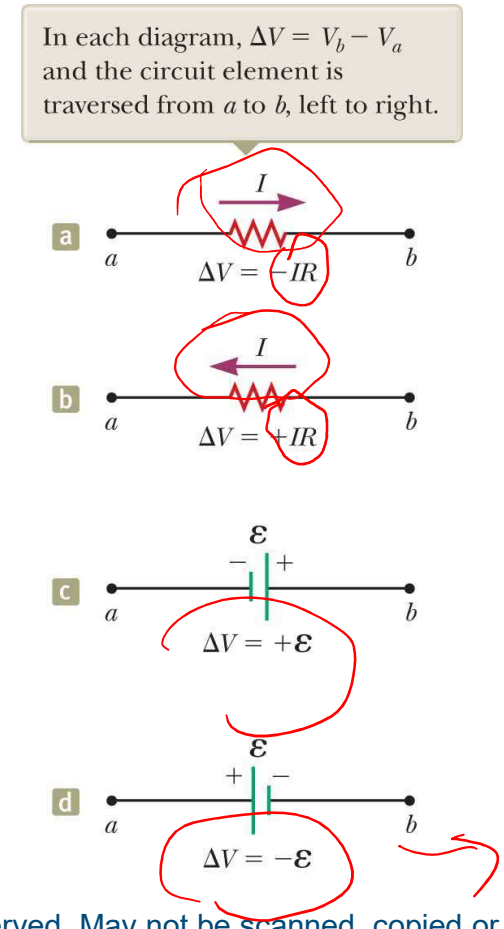


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Sign Conventions for Kirchhoff's Rules





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Problem-Solving Strategy: Kirchhoff's Rules

1. Conceptualize
2. Categorize
3. Analyze
4. Finalize

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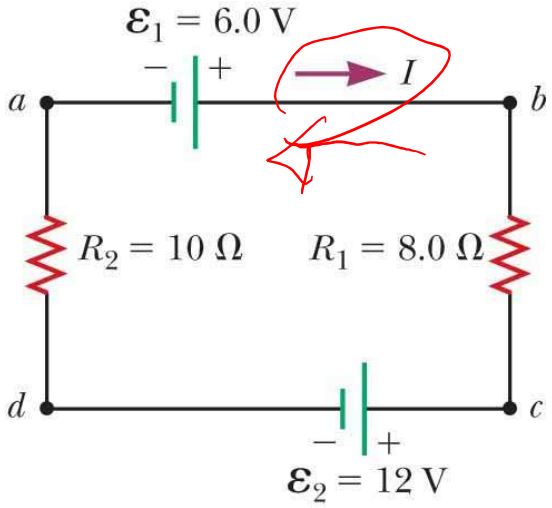
Example 27.6: A Single-Loop Circuit (1 of 5)

A single-loop circuit contains two resistors and two batteries as shown in the figure. (Neglect the internal resistances of the batteries.) Find the current in the circuit.

$$\sum \Delta V = 0 \Rightarrow \varepsilon_1 - IR_1 - \varepsilon_2 - IR_2 = 0$$

$$I = \frac{\varepsilon_1 - \varepsilon_2}{R_1 + R_2}$$

$$= \frac{6.0\ \text{V} - 12\ \text{V}}{8.0\ \Omega + 10\ \Omega} = \boxed{-0.33\ \text{A}}$$



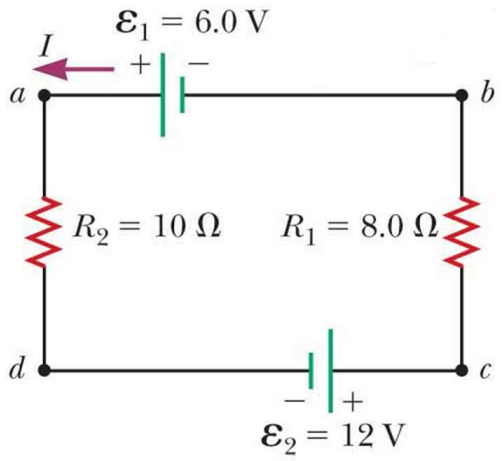
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Example 27.6: A Single-Loop Circuit (2 of 5)

What if the polarity of the 12.0-V battery were reversed? How would that affect the circuit?

$$I = \frac{\mathcal{E}_1 + \mathcal{E}_2}{R_1 + R_2} = \frac{6.0\ \text{V} + 12\ \text{V}}{8.0\ \Omega + 10\ \Omega} = \boxed{1.0\ \text{A}}$$



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Example 27.6: A Single-Loop Circuit (3 of 5)

Find the currents I_1, I_2, and I_3 in the circuit shown in the figure.

$$I_1 + I_2 - I_3 = 0$$

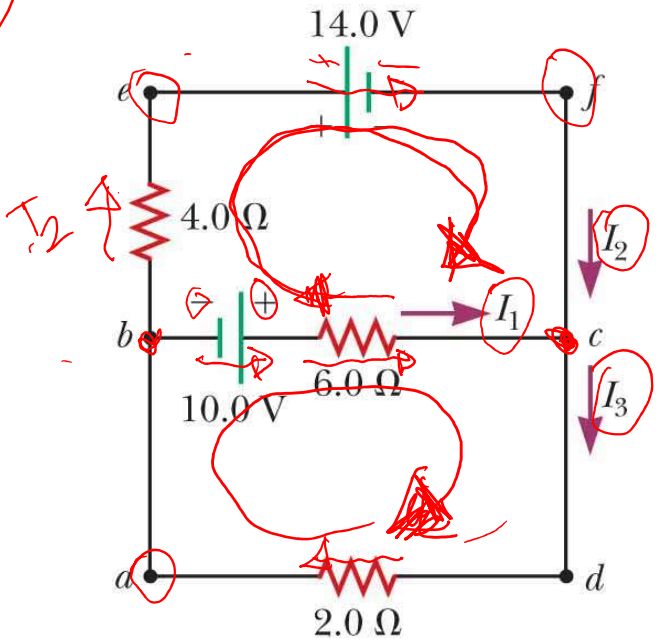
abcda:

$$+10.0\ \text{V} - (6.0\ \Omega)I_1 - (2.0\ \Omega)I_3 = 0$$

befcb:

$$-(4.0\ \Omega)I_2 - 14.0\ \text{V} + (6.0\ \Omega)I_1 - 10.0\ \text{V} = 0$$

$$-24.0\ \text{V} + (6.0\ \Omega)I_1 - (4.0\ \Omega)I_2 = 0$$



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Example 27.6: A Single-Loop Circuit (4 of 5)

$$I_1 + I_2 - I_3 = 0 \qquad \Rightarrow I_3 = I_1 + I_2$$

$$10.0\ \text{V} - (6.0\ \Omega) I_1 - (2.0\ \Omega)(I_1 + I_2) = 0$$

$$10.0\ \text{V} - (8.0\ \Omega) I_1 - (2.0\ \Omega) I_2 = 0$$

$$4\left(-24.0\ \text{V} + (6.0\ \Omega) I_1 - (4.0\ \Omega) I_2 = 0\right)$$

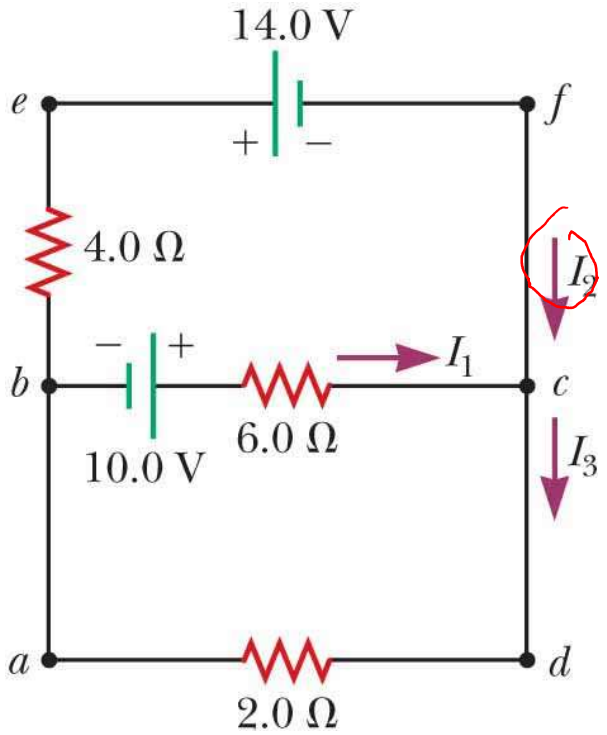
$$-96.0\ \text{V} + (24.0\ \Omega) I_1 - (16.0\ \Omega) I_2 = 0$$

$$3\left(10.0\ \text{V} - (8.0\ \Omega) I_1 - (2.0\ \Omega) I_2 = 0\right)$$

$$30.0\ \text{V} - (24.0\ \Omega) I_1 - (6.0\ \Omega) I_2 = 0$$

$$-66.0\ \text{V} - (22.0\ \Omega) I_2 = 0$$

$$I_2 = \boxed{-3.0\ \text{A}}$$



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Example 27.6: A Single-Loop Circuit (5 of 5)

$$I_2 = -3.0\ \text{A} \Rightarrow$$

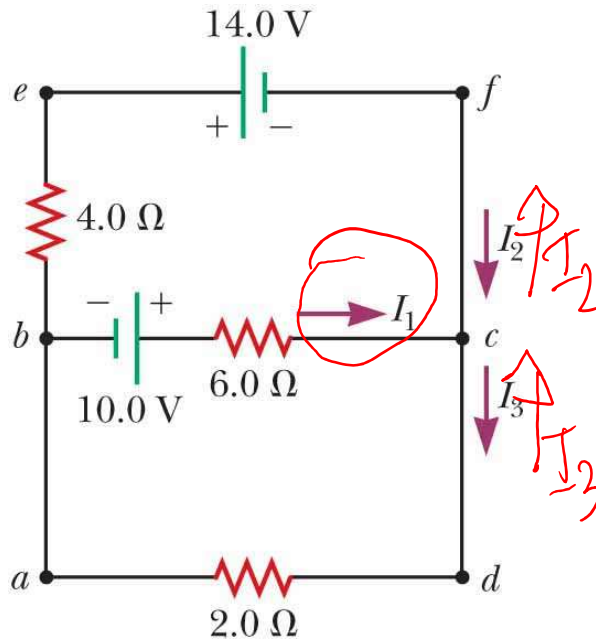
$$-24.0\ \text{V} + (6.0\ \Omega) I_1 - (4.0\ \Omega)(-3.0\ \text{A}) = 0$$

$$-24.0\ \text{V} + (6.0\ \Omega) I_1 + 12.0\ \text{V} = 0$$

$$I_1 = \boxed{2.0\ \text{A}}$$

$$I_3 = I_1 + I_2 \Rightarrow$$

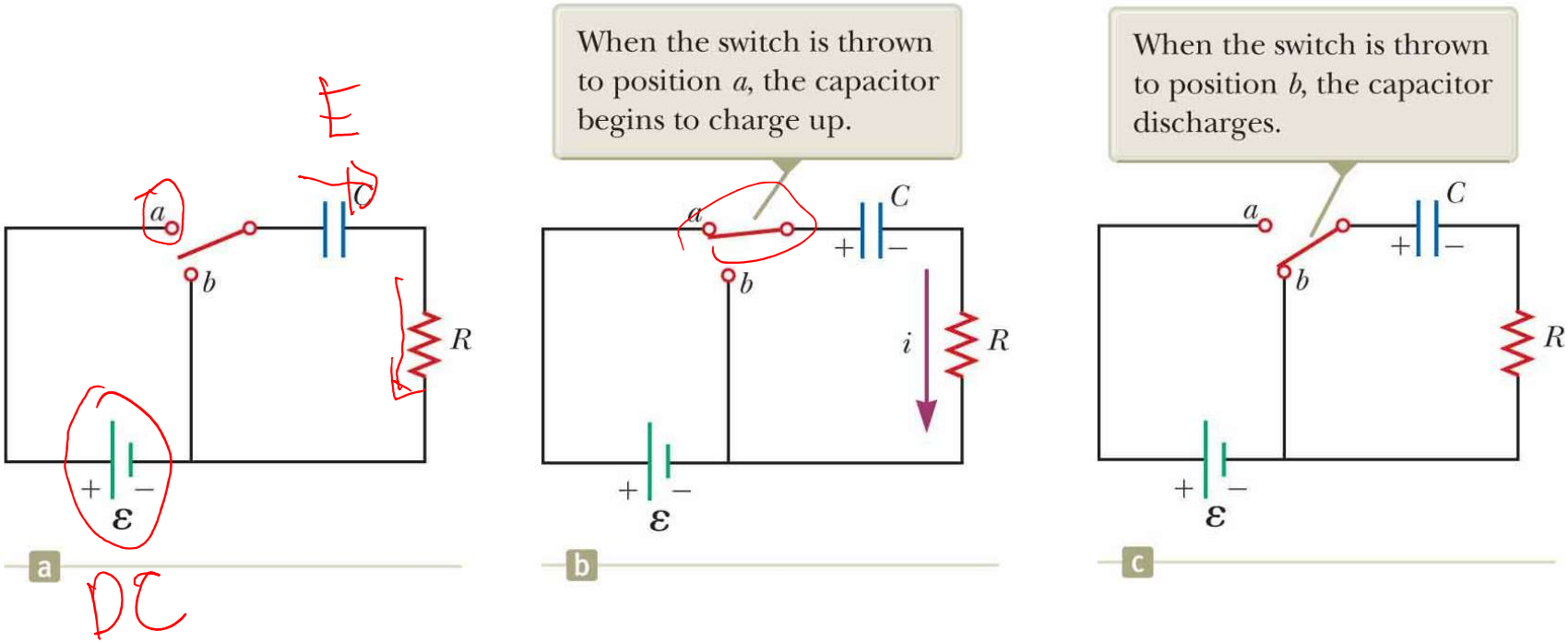
$$I_3 = 2.0\ \text{A} - 3.0\ \text{A} = \boxed{-1.0\ \text{A}}$$



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Charging a Capacitor (1 of 5)



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Charging a Capacitor (2 of 5)

$$\varepsilon - \frac{q}{C} - iR = 0$$

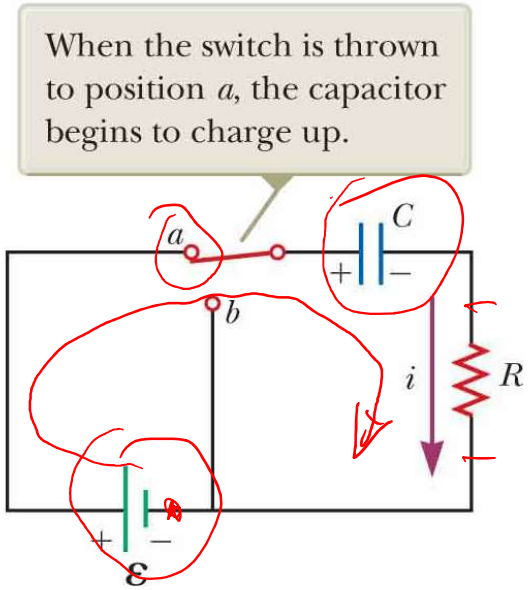
$$I_i = \frac{\varepsilon}{R} \quad \text{(current at } t = 0\text{)}$$

$$Q_{\max} = C\varepsilon \quad \text{(maximum charge)}$$

$$\frac{dq}{dt} = \frac{\varepsilon}{R} - \frac{q}{RC}$$

$$\frac{dq}{dt} = \frac{C\varepsilon}{RC} - \frac{q}{RC} = -\frac{q - C\varepsilon}{RC}$$

$$\frac{dq}{q - C\varepsilon} = -\frac{1}{RC}dt$$



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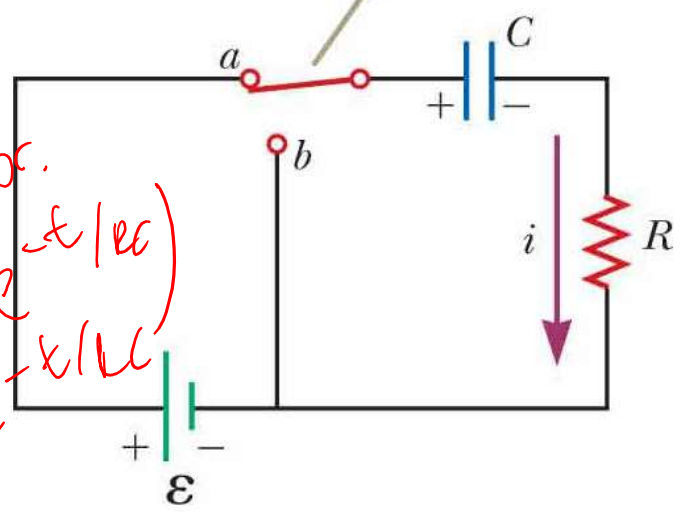
Charging a Capacitor (3 of 5)

$$\int_0^q \frac{dq}{q - C\varepsilon} = -\frac{1}{RC}\int_0^t dt \Rightarrow \ln\left(\frac{q - C\varepsilon}{-C\varepsilon}\right) = -\frac{t}{RC}$$

$$q(t) = C\varepsilon\left(1 - e^{-t/RC}\right)$$
$$= Q_{\max}\left(1 - e^{-t/RC}\right)$$

$$i(t) = \frac{\varepsilon}{R}e^{-t/RC}$$

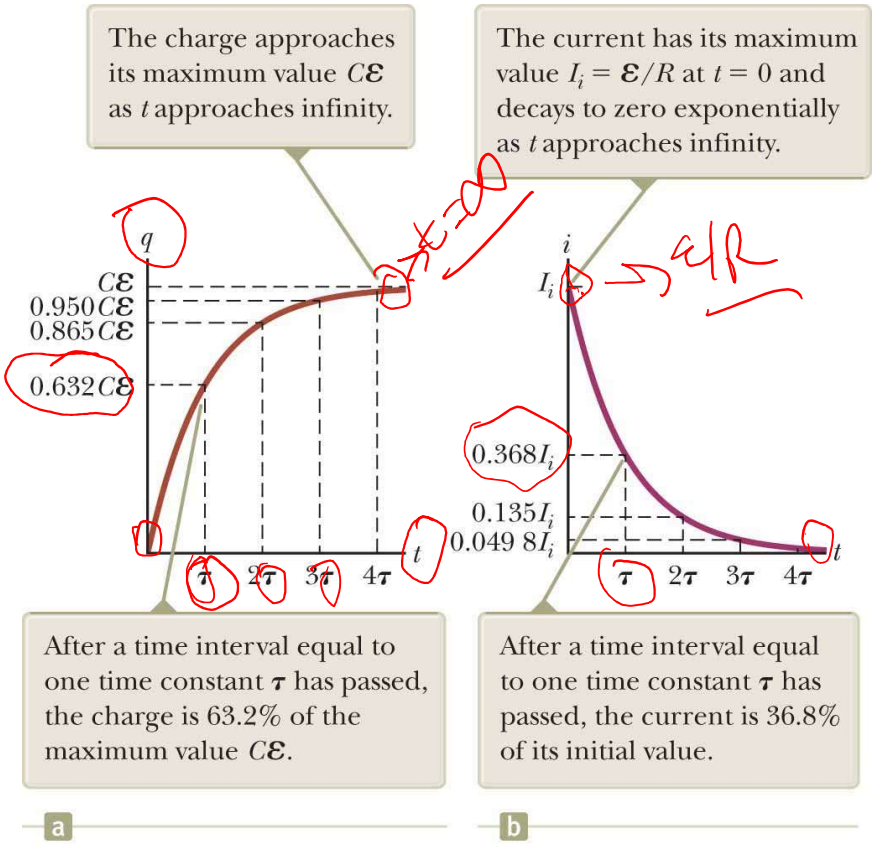
When the switch is thrown to position *a*, the capacitor begins to charge up.



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Charging a Capacitor (4 of 5)



time constant

$$\tau = RC$$

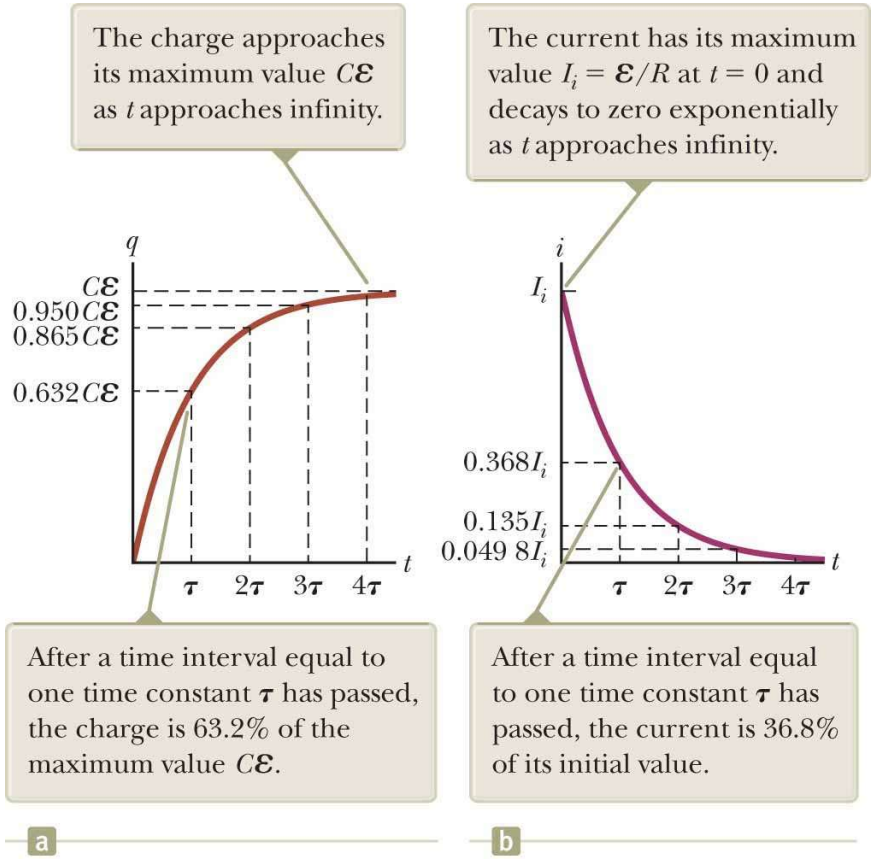
$$i = e^{-1} I_i = 0.368 I_i$$

$$i = e^{-2} I_i = 0.135 I_i$$

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Charging a Capacitor (5 of 5)



$$C\varepsilon\left[1-e^{-1}\right]=0.632C\varepsilon$$

$$Q_{\max}\varepsilon = C\varepsilon^2$$

$$\frac{1}{2}Q_{\max}\varepsilon = \frac{1}{2}C\varepsilon^2$$

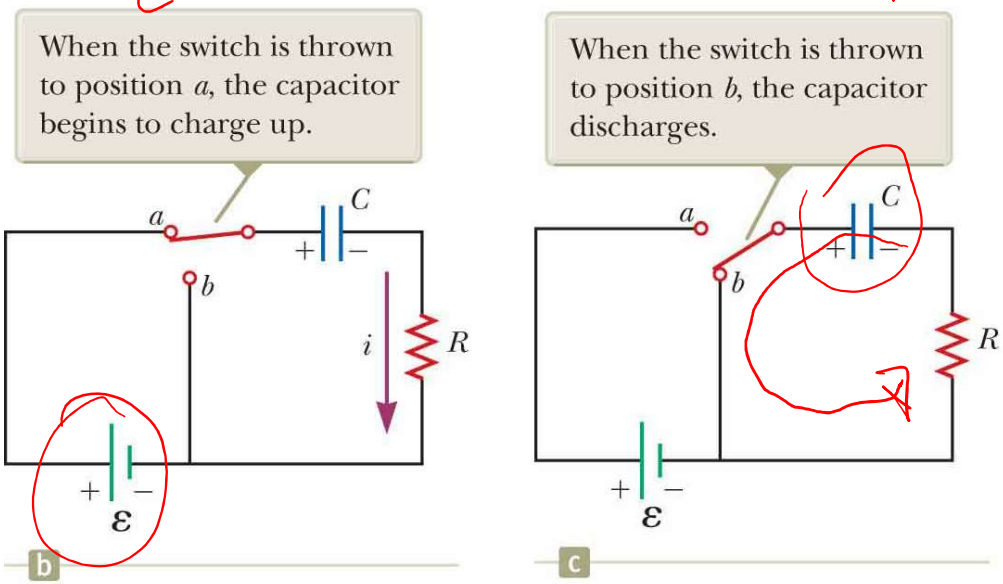
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Discharging a Capacitor (1 of 2)

$$-\frac{q}{C} - iR = 0$$

$$-R\frac{dq}{dt} = \frac{q}{C} \Rightarrow \frac{dq}{q} = -\frac{1}{RC}dt$$



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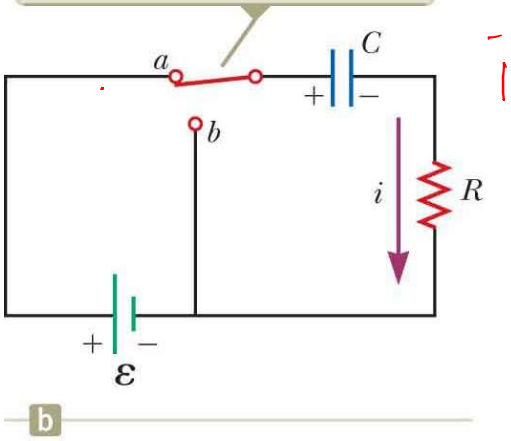
Discharging a Capacitor (2 of 2)

$$\int_{Q_i}^{q} \frac{dq}{q} = -\frac{1}{RC}\int_0^t dt \Rightarrow \ln\left(\frac{q}{Q_i}\right) = -\frac{t}{RC}$$

$$q(t) = Q_i e^{-t/RC}$$

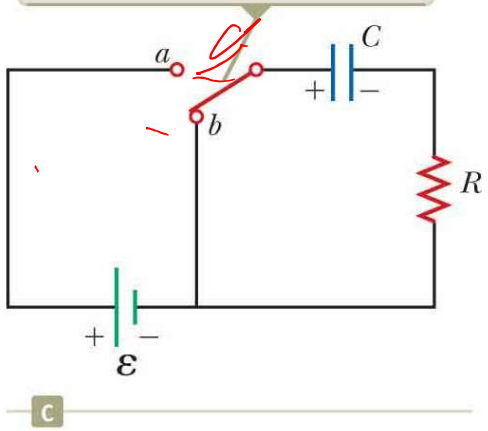
$$i(t) = -\frac{Q_i}{RC} e^{-t/RC}$$

When the switch is thrown to position *a*, the capacitor begins to charge up.



b

When the switch is thrown to position *b*, the capacitor discharges.



c

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Quick Quiz 27.5 Part I (1 of 2)

Consider the circuit in the figure and assume the battery has no internal resistance. Just after the switch is closed, what is the current in the battery?

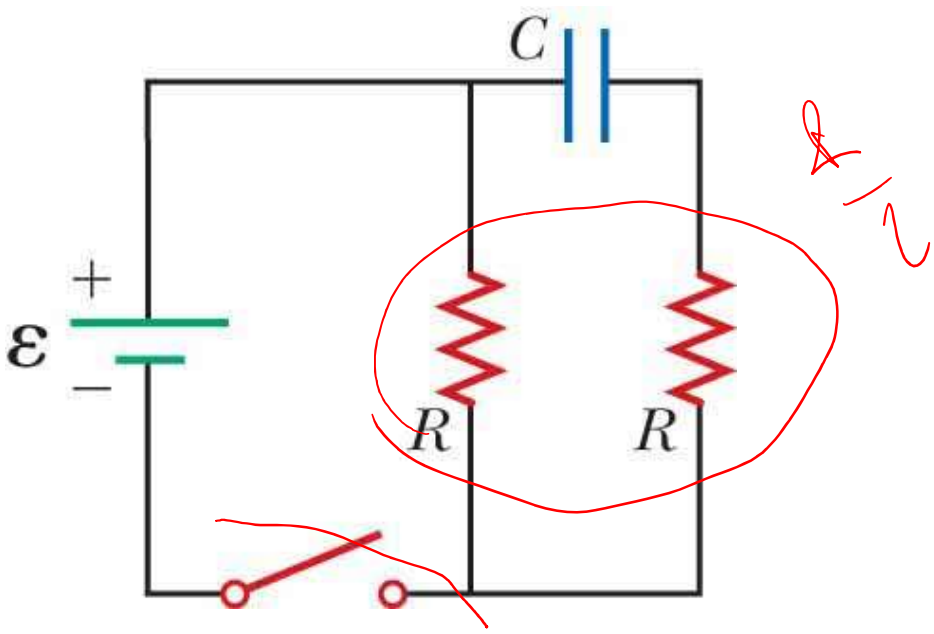
(a) 0

(b) $\mathcal{E}/2R$

(c) $2\mathcal{E}/R$

(d) $\mathcal{E}/R$

(e) impossible to determine



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Quick Quiz 27.5 Part I (2 of 2)

Consider the circuit in the figure and assume the battery has no internal resistance. Just after the switch is closed, what is the current in the battery?

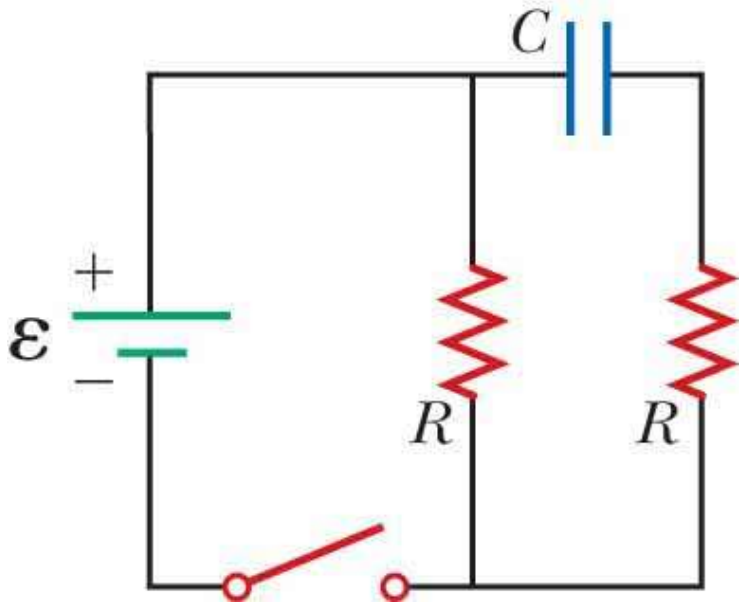
(a) 0

(b) $\mathcal{E}/2R$

(c) $2\mathcal{E}/R$

(d) $\mathcal{E}/R$

(e) impossible to determine



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Quick Quiz 27.5 Part II (1 of 2)

After a very long time, what is the current in the battery?

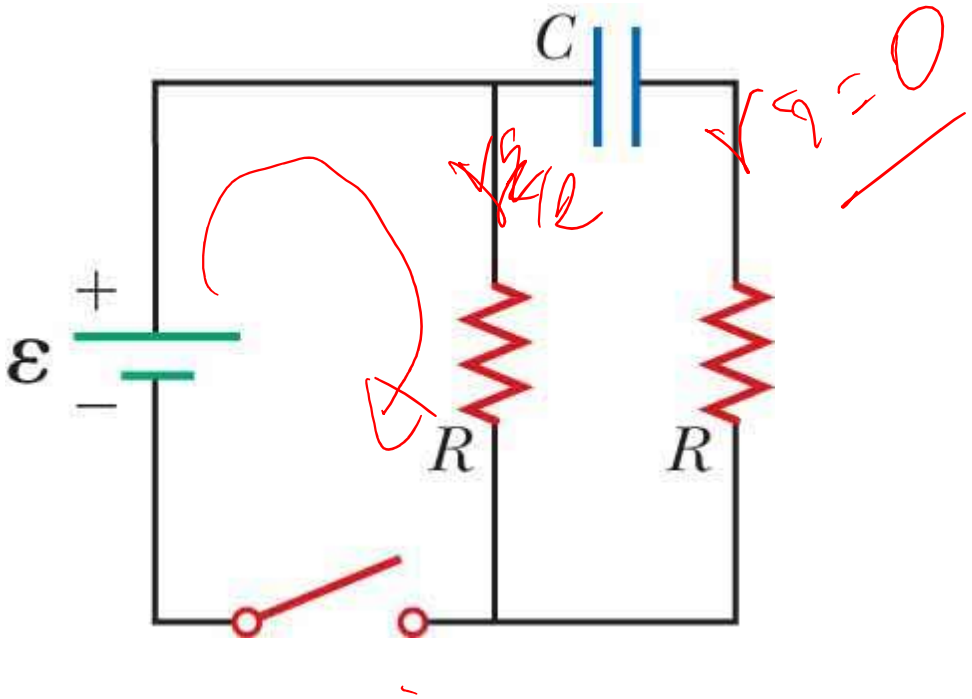
(a) 0

(b) $\varepsilon/2R$

(c) $2\varepsilon/R$

(d) ε/R

(e) impossible to determine



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Quick Quiz 27.5 Part II (2 of 2)

After a very long time, what is the current in the battery?

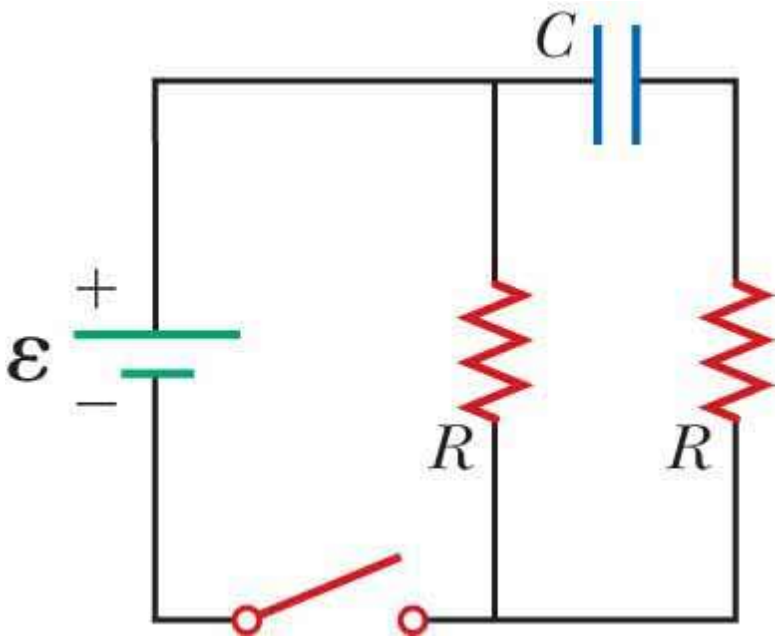
(a) 0

(b) $\varepsilon/2R$

(c) $2\varepsilon/R$

(d) $\boldsymbol{\varepsilon/R}$

(e) impossible to determine



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Conceptual Example 27.8: Intermittent Windshield Wipers

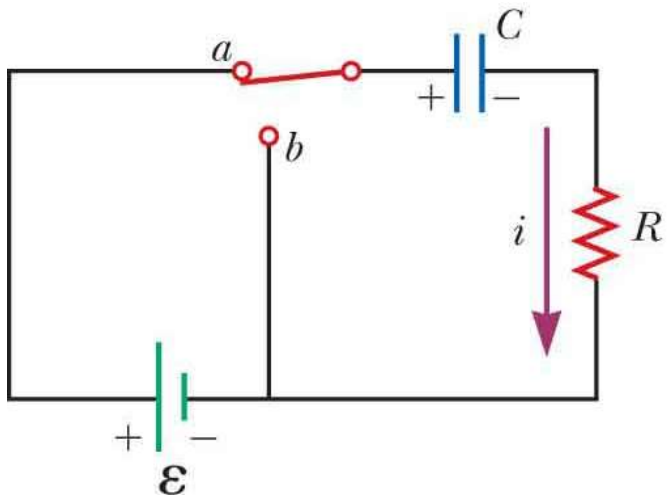
Many automobiles are equipped with windshield wipers that can operate intermittently during a light rainfall. How does the operation of such wipers depend on the charging and discharging of a capacitor?

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Example 27.9: Charging a Capacitor in an RC Circuit (1 of 2)

An uncharged capacitor and a resistor are connected in series to a battery as shown in the figure, where $\varepsilon = 12.0$ V, $C = 5.00\ \mu$F, and $R = 8.00 \times 10^5\ \Omega$. The switch is thrown to position *a*. Find the time constant of the circuit, the maximum charge on the capacitor, the maximum current in the circuit, and the charge and current as functions of time.



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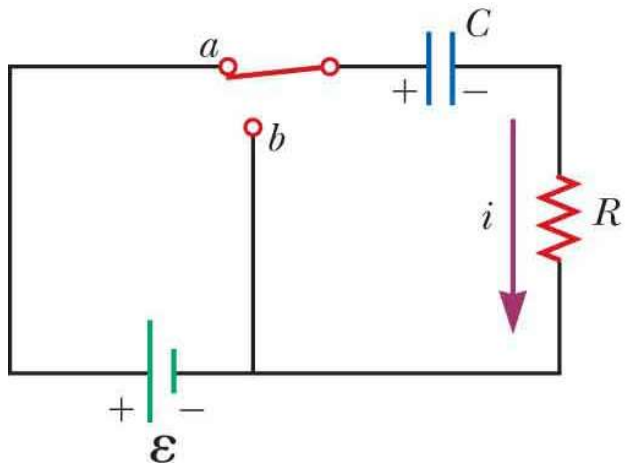
Example 27.9: Charging a Capacitor in an RC Circuit (2 of 2)

$$\tau = RC = \left(8.00\times10^{5}\ \Omega\right)\left(5.00\times10^{-6}\ \text{F}\right) = \boxed{4.00\ \text{s}}$$

$$Q_{\max} = C\varepsilon = \left(5.00\ \mu\text{F}\right)\left(12.0\ \text{V}\right) = \boxed{60.0\ \mu\text{C}}$$

$$I_i = \frac{\varepsilon}{R} = \frac{12.0\ \text{V}}{8.00\times10^{5}\ \Omega} = \boxed{15.0\ \mu\text{A}}$$

$$q(t) = 60.0\left(1-e^{-t/4.00}\right)$$
$$i(t) = 15.0e^{-t/4.00}$$



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Example 27.10: Discharging a Capacitor in an RC Circuit (1 of 3)

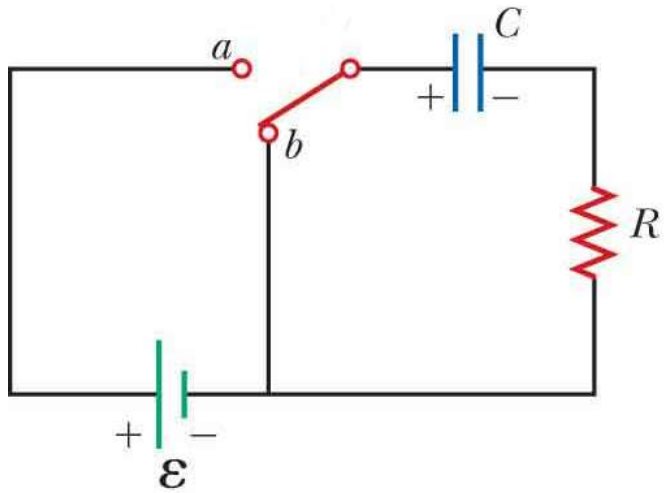
Consider a capacitor of capacitance *C* that is being discharged through a resistor of resistance *R* as shown in the figure.

(A) After how many time constants is the charge on the capacitor one-fourth its initial value?

$$\frac{Q_i}{4} = Q_i e^{-t/RC} \Rightarrow \frac{1}{4} = e^{-t/RC}$$

$$-\ln 4 = -\frac{t}{RC}$$

$$t = RC \ln 4 = 1.39RC = \boxed{1.39\tau}$$



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Example 27.10: Discharging a Capacitor in an RC Circuit (2 of 3)

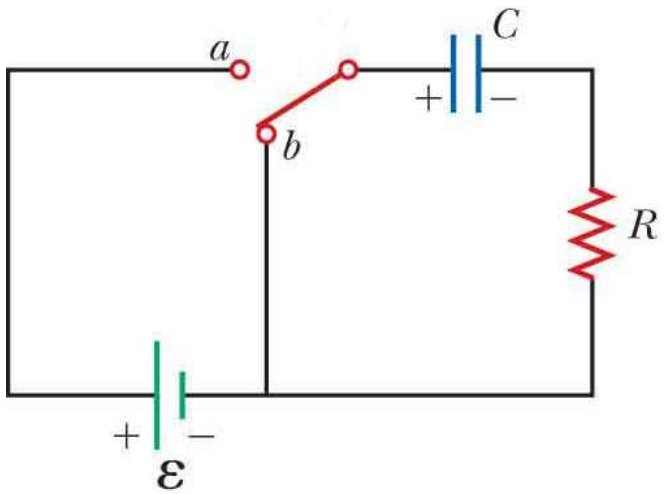
(B) The energy stored in the capacitor decreases with time as the capacitor discharges. After how many time constants is this stored energy one-fourth its initial value?

$$U(t) = \frac{q^2}{2C} = \frac{Q_i^2}{2C} e^{-2t/RC}$$

$$\frac{1}{4}\frac{Q_i^2}{2C} = \frac{Q_i^2}{2C} e^{-2t/RC} \Rightarrow \frac{1}{4} = e^{-2t/RC}$$

$$-\ln 4 = -\frac{2t}{RC}$$

$$t = \frac{1}{2} RC \ln 4 = 0.693RC = 0.693\tau$$



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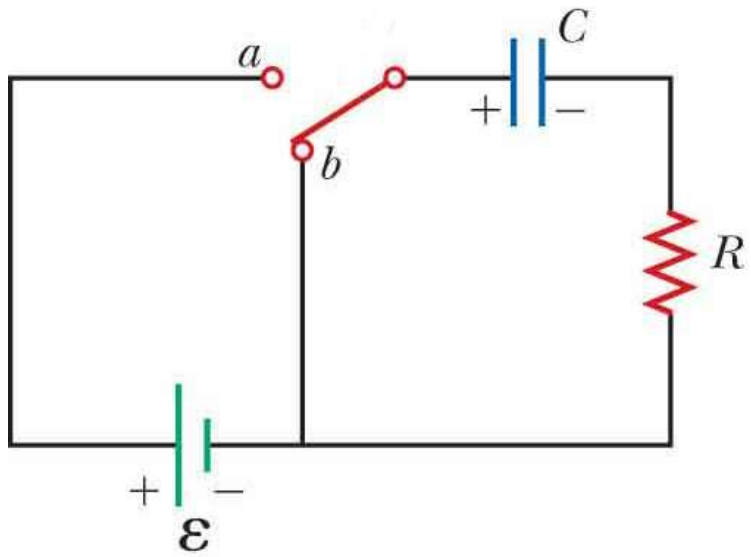
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Example 27.10: Discharging a Capacitor in an RC Circuit (3 of 3)

What if you want to describe the circuit in terms of the time interval required for the charge to fall to one-half its original value rather than by the time constant τ? That would give a parameter for the circuit called its half-life $\tau_{1/2}$. How is the half-life related to the time constant?

$$\frac{Q_i}{2} = Q_i e^{-t_{1/2}/RC} \Rightarrow \frac{1}{2} = e^{-t_{1/2}/RC}$$

$$t_{1/2} = 0.693\tau$$



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Example 27.11: Energy Delivered to a Resistor (1 of 2)

A 5.00-μF capacitor is charged to a potential difference of 800 V and then discharged through a resistor. How much energy is delivered to the resistor in the time interval required to fully discharge the capacitor?

$$\Delta U + \Delta E_{\text{int}} = 0$$

$$\left(0 - U_E\right) + \left(E_{\text{int}} - 0\right) = 0 \Rightarrow E_R = U_E$$

$$E_R = \frac{1}{2} C \varepsilon^2$$

$$E_R = \frac{1}{2}\left(5.00 \times 10^{-6}\ \text{F}\right)\left(800\ \text{V}\right)^2 = \boxed{1.60\ \text{J}}$$

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Example 27.11: Energy Delivered to a Resistor (2 of 2)

$$i(t) = -\frac{Q_i}{RC} e^{-t/RC}$$

$$P = \frac{dE}{dt} \Rightarrow E_R = \int_0^\infty P\, dt$$

$$E_R = \int_0^\infty i^2 R\, dt$$

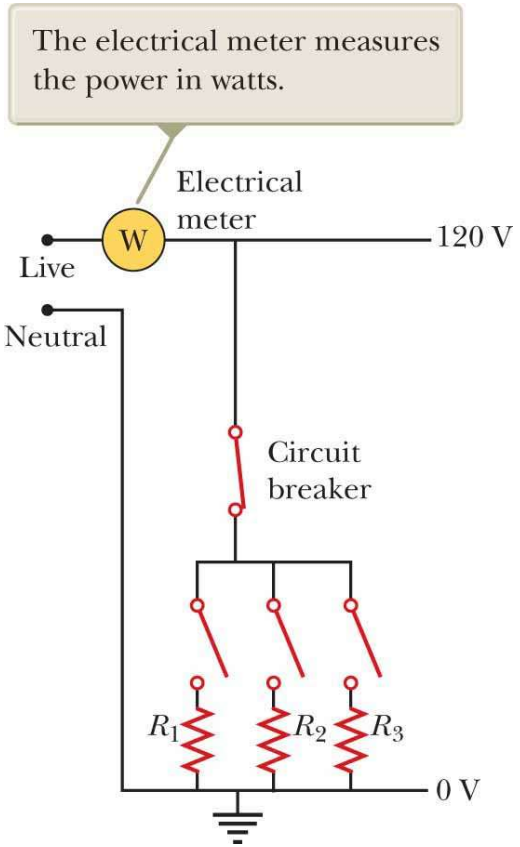
$$E_R = \int_0^\infty \left(-\frac{Q_i}{RC} e^{-t/RC} \right)^2 R\, dt = \frac{Q_i^2}{RC^2} \int_0^\infty e^{-2t/RC}\, dt = \frac{\varepsilon^2}{R} \int_0^\infty e^{-2t/RC}\, dt$$

$$E_R = \frac{\varepsilon^2}{R} \left(\frac{RC}{2} \right) = \frac{1}{2} C \varepsilon^2$$

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Household Wiring (1 of 2)



$$P = IV \rightarrow I = \frac{P}{V}$$

$$I_{\text{toaster}} = \frac{1000\ \text{W}}{120\ \text{V}} = 8.33\ \text{A}$$

$$I_{\text{microwave}} = \frac{1300\ \text{W}}{120\ \text{V}} = 10.8\ \text{A}$$

$$I_{\text{coffee maker}} = \frac{800\ \text{W}}{120\ \text{V}} = 6.67\ \text{A}$$

$$I_{\text{total}} = 8.33\ \text{A} + 10.8\ \text{A} + 6.67\ \text{A} = 25.8\ \text{A}$$

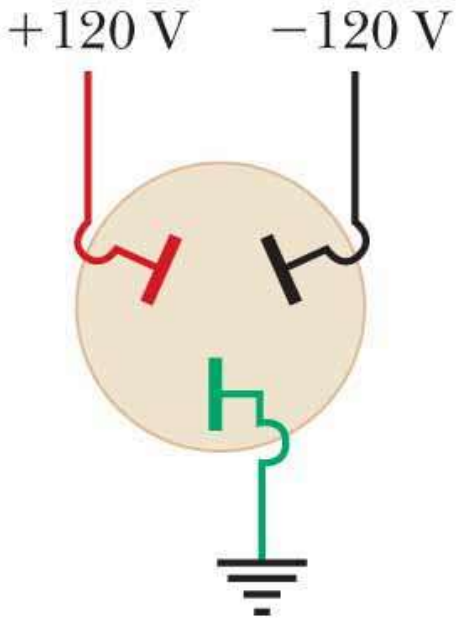
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Household Wiring (2 of 2)



a

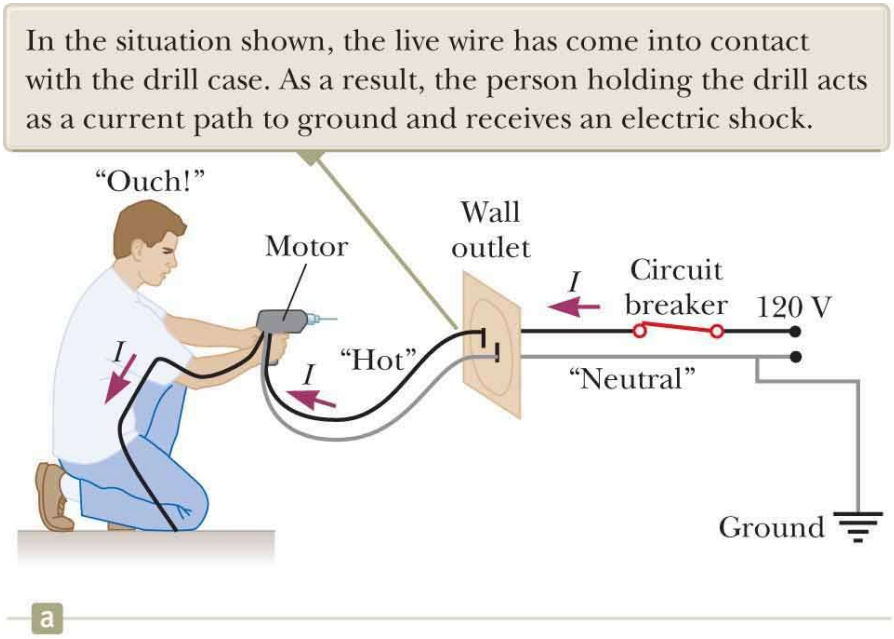


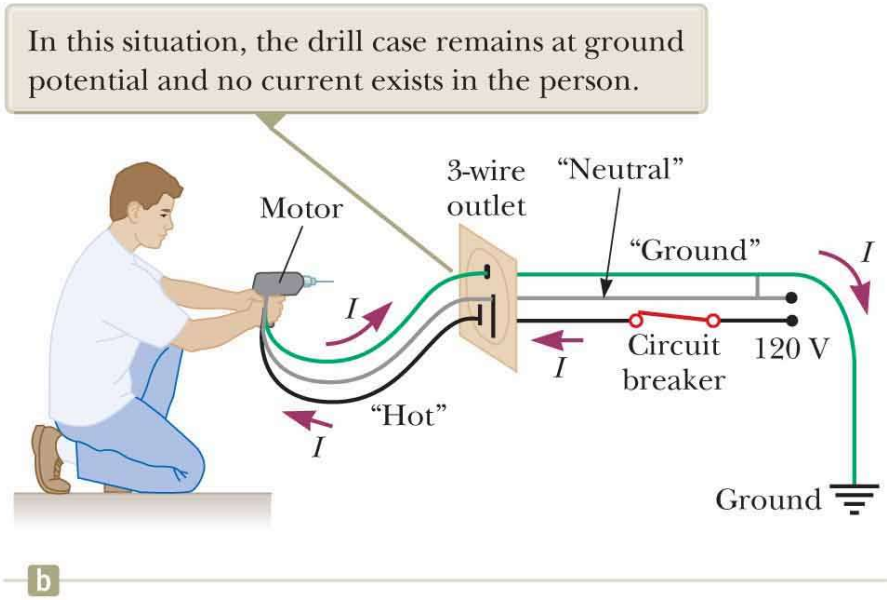
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Electrical Safety (1 of 2)





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Electrical Safety (2 of 2)

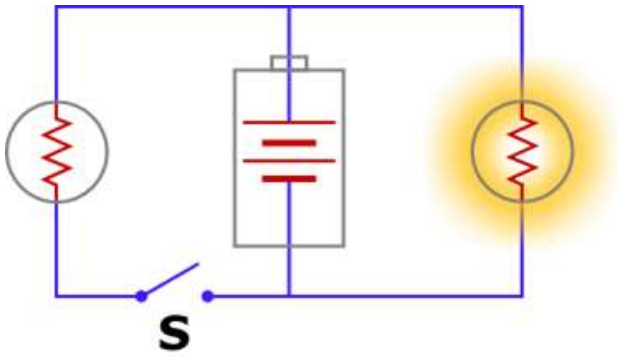


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Assessing to Learn (1 of 21)

A battery is used to light a bulb as shown. A second bulb is connected by closing switch S. What happens to the bulbs?

1. The right bulb goes out; the left bulb lights up.
2. The right bulb gets noticeably dimmer; both bulbs have the same brightness.
3. The right bulb gets very slightly dimmer; both bulbs have the same brightness.
4. The right bulb's brightness stays the same; both bulbs have the same brightness.
5. The right bulb gets slightly brighter; both bulbs have the same brightness.
6. The right bulb gets noticeably brighter; both bulbs have the same brightness.
7. The right bulb stays the same; the left bulb does not light.
8. Impossible to determine



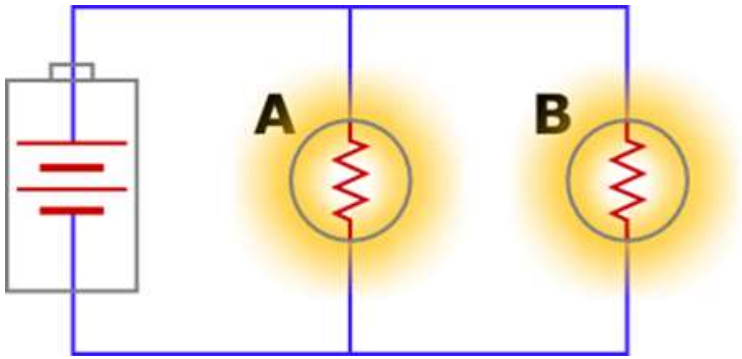
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Assessing to Learn (2 of 21)

A battery is used to light two bulbs as shown. What happens to bulb B when bulb A is unscrewed from its socket?

1. Bulb B goes out.
2. Bulb B gets noticeably dimmer, without going out.
3. Bulb B gets slightly dimmer.
4. Bulb B's brightness stays the same.
5. Bulb B gets slightly brighter.
6. Bulb B gets noticeably brighter.
7. Impossible to determine



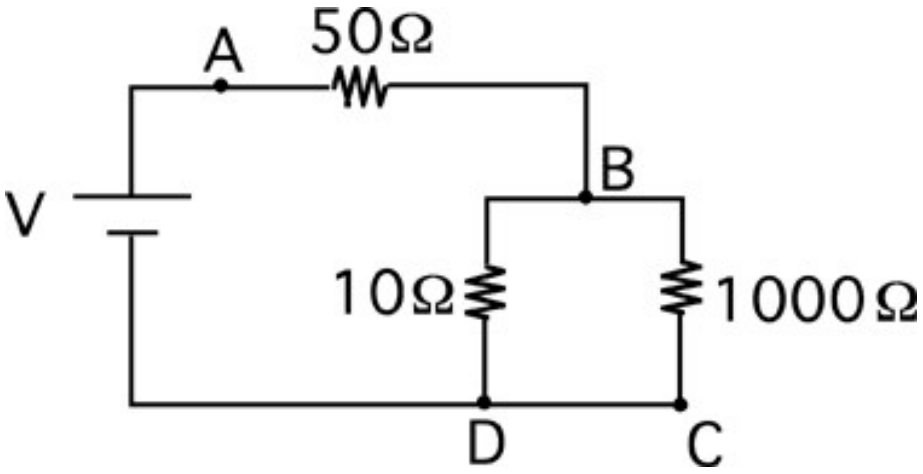
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Assessing to Learn (3 of 21)

Consider the circuit below. Which resistor has the largest power consumption?

1. The 50-Ω resistor
2. The 10-Ω resistor
3. The 1000-Ω resistor
4. The 50-Ω and 10-Ω resistors
5. The 50-Ω and 1000-Ω resistors
6. The 10-Ω and 1000-Ω resistors
7. All have the same power.



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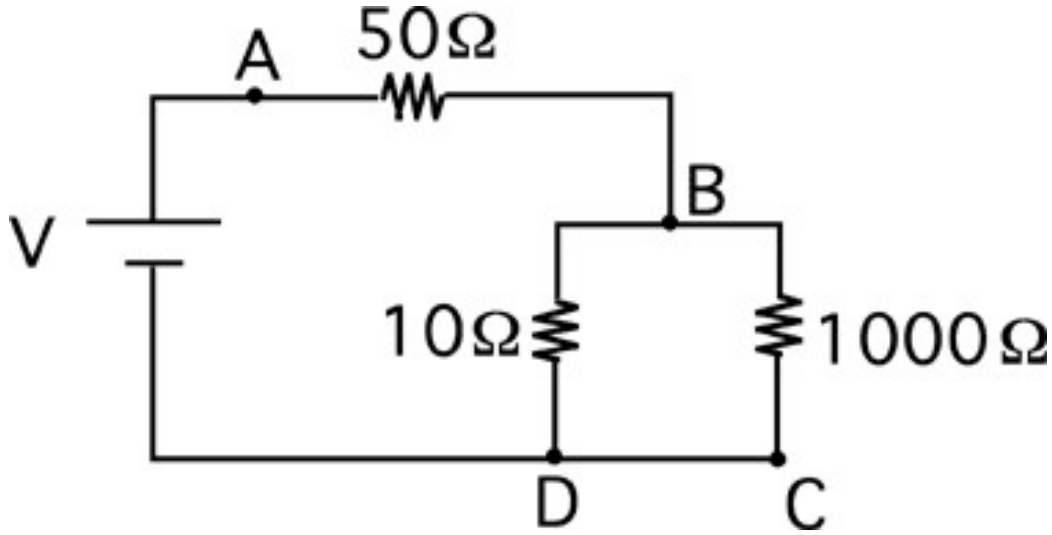
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Assessing to Learn (4 of 21)

Consider the circuit below. Which statement(s) is correct?

1. $I_{AB} = I_{BD} + I_{BC}$
2. $I_{BD} > I_{BC}$
3. $I_{BD} < I_{BC}$

4. 1 only
5. 2 only
6. 3 only
7. 1 and 2
8. 1 and 3



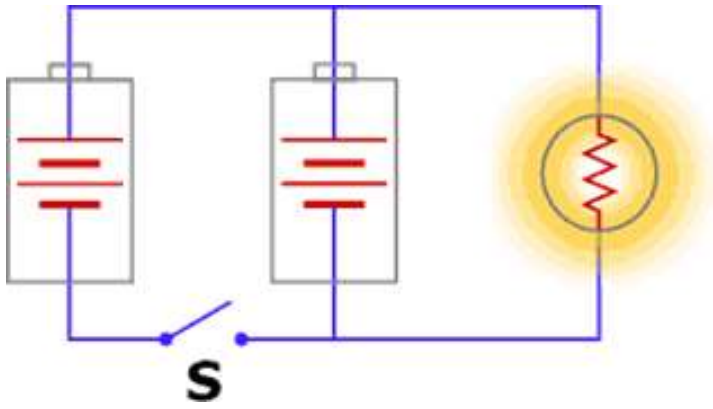
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Assessing to Learn (5 of 21)

A battery is used to light a bulb as shown. An identical battery is connected by closing switch S. What happens to the bulb?

1. The bulb goes out.
2. The bulb gets noticeably dimmer.
3. The bulb gets very slightly dimmer.
4. The bulb's brightness stays the same.
5. The bulb gets slightly brighter.
6. The bulb gets noticeably brighter.
7. A combination of 2 of the answers above; first one happens, then the other.



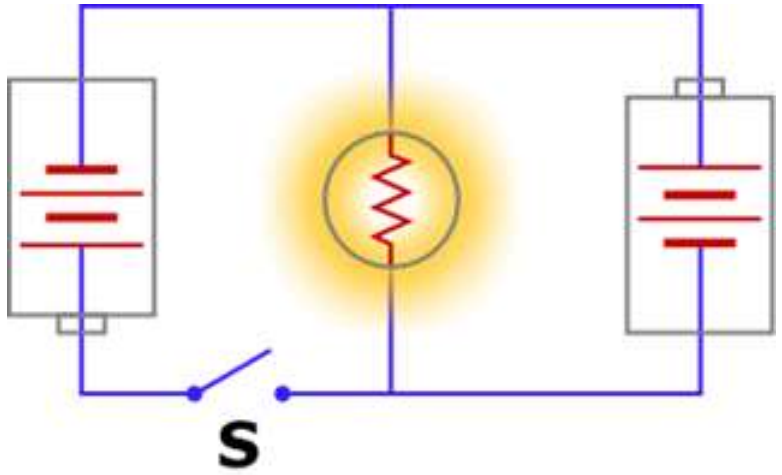
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Assessing to Learn (6 of 21)

A battery is used to light a bulb as shown. An identical battery is connected by closing switch S. What happens to the bulb?

1. The bulb goes out.
2. The bulb gets noticeably dimmer.
3. The bulb gets very slightly dimmer.
4. The bulb's brightness stays the same.
5. The bulb gets slightly brighter.
6. The bulb gets noticeably brighter.
7. Impossible to determine



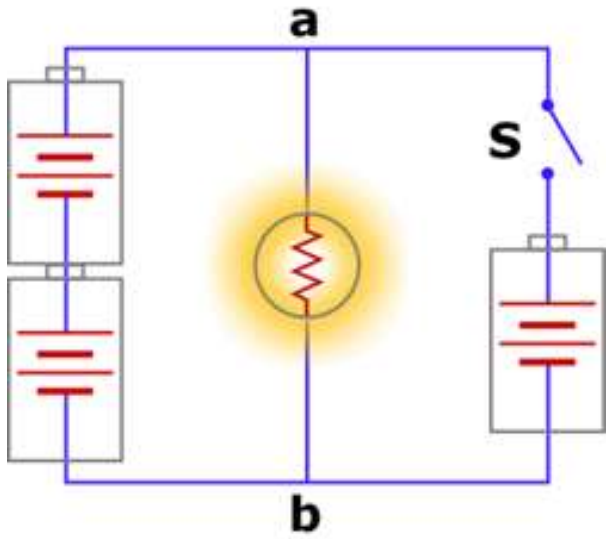
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Assessing to Learn (7 of 21)

Two batteries are used to light a bulb as shown. A third identical battery is connected by closing switch S. What happens to the bulb?

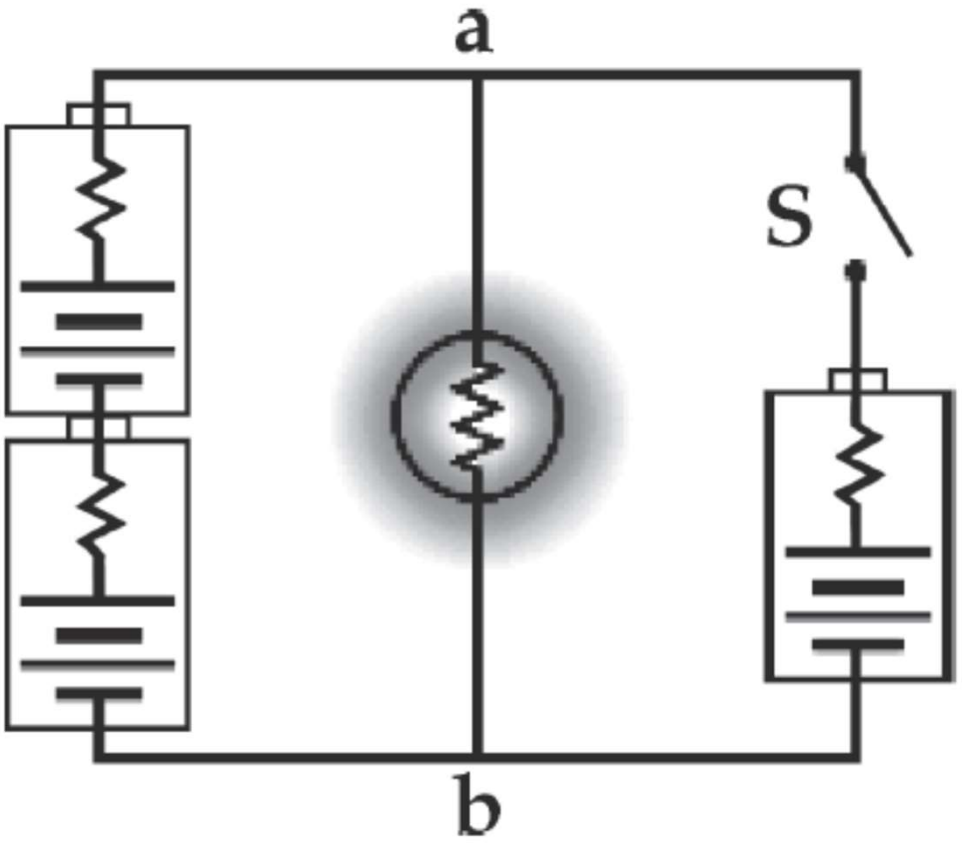
1. The bulb goes out.
2. The bulb gets noticeably dimmer.
3. The bulb gets very slightly dimmer.
4. The bulb's brightness stays the same.
5. The bulb gets slightly brighter.
6. The bulb gets noticeably brighter.
7. Impossible to determine



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Assessing to Learn (8 of 21)



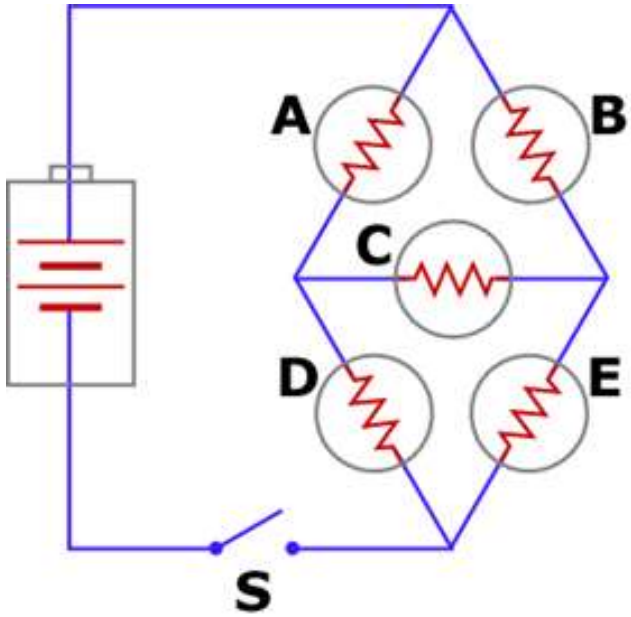
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Assessing to Learn (9 of 21)

A battery and 5 identical bulbs are arranged as shown. What happens when switch S is closed?

1. All 5 bulbs turn on with equal brightness.
2. All 5 bulbs turn on, but they do not all have equal brightness.
3. One of the bulbs does not turn on; the other 4 turn on with equal brightness.
4. One of the bulbs does not turn on; the other 4 turn on, but they do not all have equal brightness.
5. Two or more bulbs do not turn on.
6. I have no idea



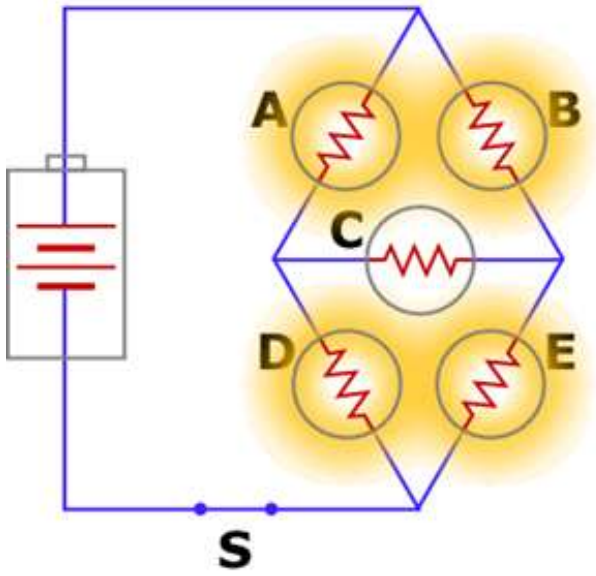
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Assessing to Learn (10 of 21)

A battery and 5 bulbs are arranged as shown, and when the switch is closed 4 of the bulbs are lit. Which bulbs change when bulb B is unscrewed from its socket?

1. Bulb E goes out. (Bulbs A and D stay on; bulb C stays off.)
2. Bulb E goes out; bulb C goes on. (Bulbs A and D stay on.)
3. Bulb A goes out; bulb C goes on. (Bulbs D and E stay on.)
4. Bulb C goes on. (Bulbs A, D, and E stay on.)
5. Bulb C goes on; bulb D goes out. (Bulbs A and E stay on.)
6. I have no idea!



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Assessing to Learn (11 of 21)

A light bulb is connected to a battery as shown in Figure A below. When a second bulb is connected as shown in Figure B, what happens to the brightness of the original bulb?

1. Increases
2. Decreases
3. Stays the same
4. Bulb goes out
5. Can't determine

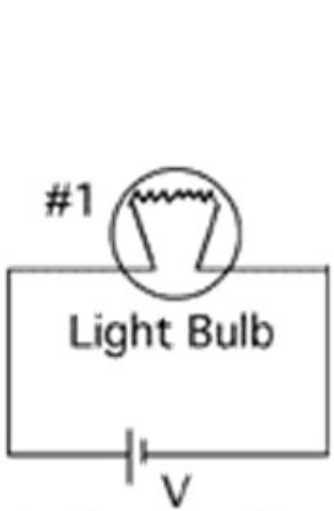


Fig. A

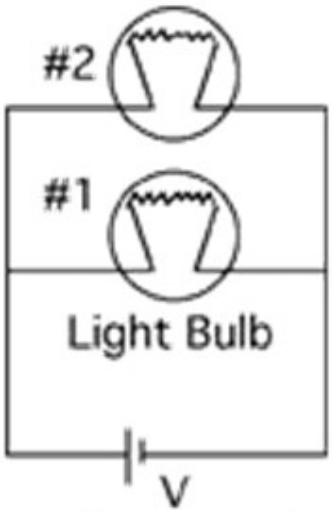


Fig. B

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Assessing to Learn (12 of 21)

A light bulb is connected to a battery as shown in Figure A below. When a second bulb is connected as shown in Figure B, what happens to the brightness of the original bulb?

1. Increases
2. Decreases
3. Stays the same
4. Bulb goes out
5. Can't determine

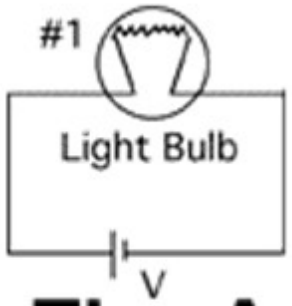


Fig. A

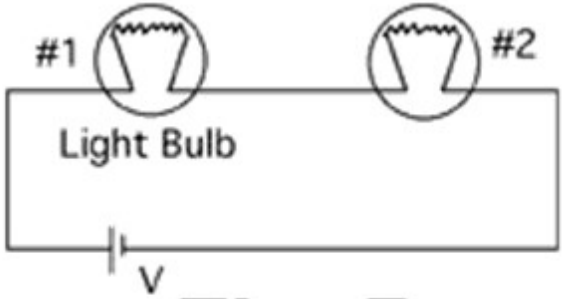


Fig. B

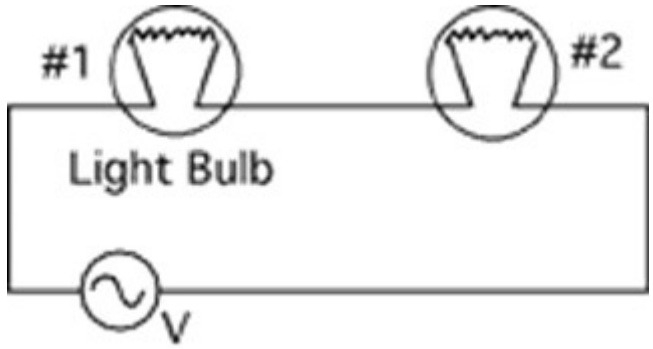
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Assessing to Learn (13 of 21)

Two light bulbs are connected to a wall outlet as shown below. Bulb #1 is 100 W and Bulb #2 is 40 W. Which statement is true?

1. Both bulbs are at their normal brightness.
2. The 100-W bulb is brighter than the 40-W bulb.
3. The 40-W bulb is brighter than the 100-W bulb.
4. Both bulbs are at equal brightness.
5. Cannot determine their relative brightness.



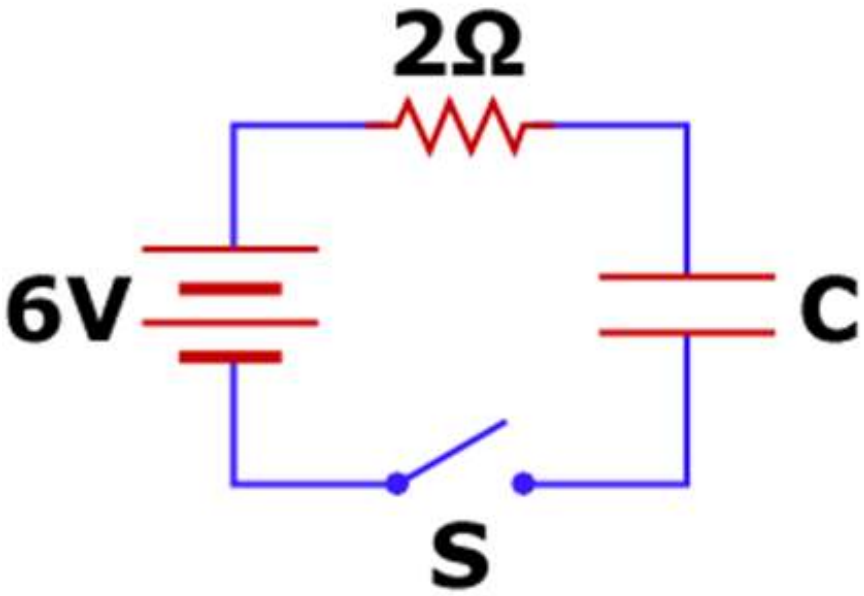
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Assessing to Learn (14 of 21)

Consider the circuit below. Switch S is closed at $t = 0$. What is the voltage across the capacitor C just after the switch is closed?

1. 0
2. 1 V
3. 2 V
4. 3 V
5. 4 V
6. 5 V
7. 6 V
8. 9 V
9. None of the above
10. Impossible to determine



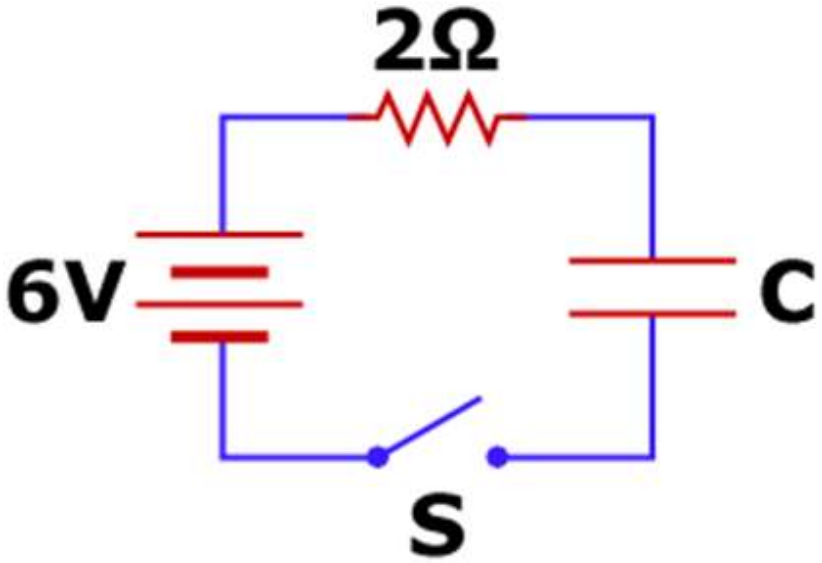
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Assessing to Learn (15 of 21)

Consider the circuit below. Switch S is closed at $t = 0$. What is the current through the capacitor C just after the switch is closed?

1. 0
2. 1 A
3. 2 A
4. 3 A
5. 4 A
6. 5 A
7. 6 A
8. 9 A
9. None of the above
10. Impossible to determine



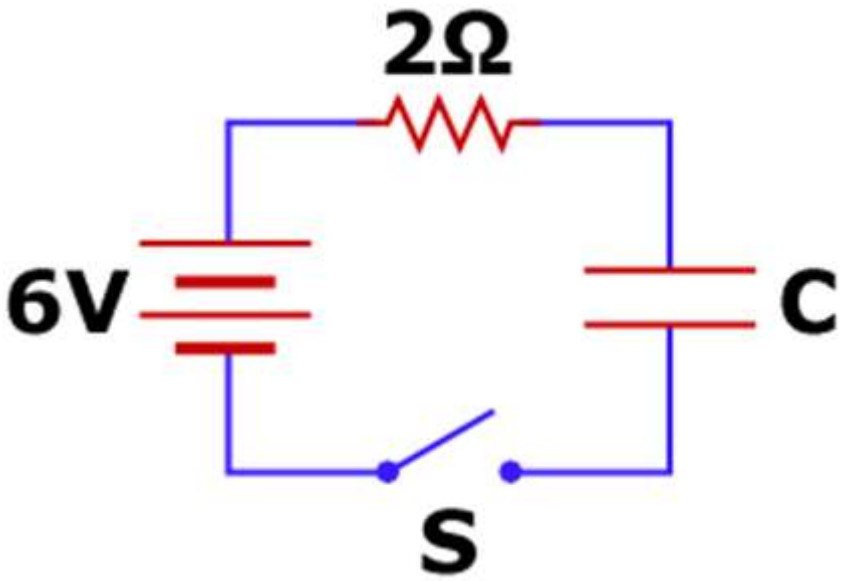
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Assessing to Learn (16 of 21)

Consider the circuit below. Switch S is closed at $t = 0$. What is the voltage across the capacitor C long after the switch is closed?

1. 0
2. 1 V
3. 2 V
4. 3 V
5. 4 V
6. 5 V
7. 6 V
8. 9 V
9. None of the above
10. Impossible to determine

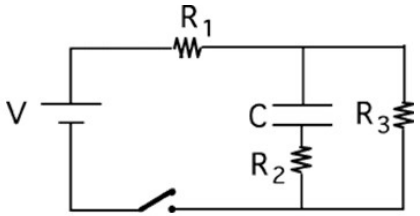


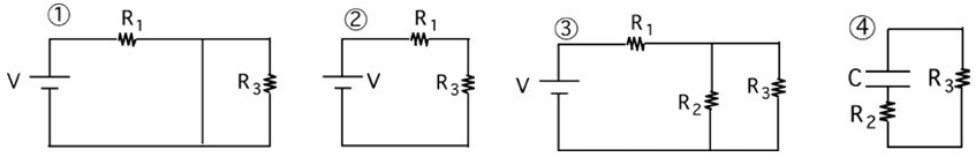
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Assessing to Learn (17 of 21)

Consider the following circuit. The capacitor is uncharged when the switch is closed at $t = 0$. Which circuit is equivalent to this circuit for the instant immediately after the switch is closed?





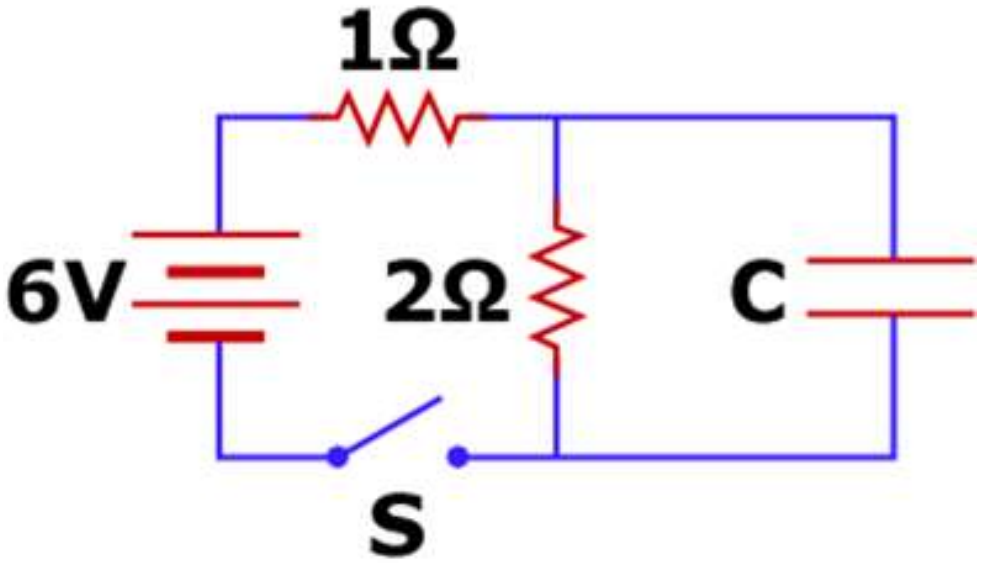
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Assessing to Learn (18 of 21)

Consider the circuit below. Switch S is closed at $t = 0$. What is the voltage across the capacitor C just after the switch is closed?

1. 0
2. 1 V
3. 2 V
4. 3 V
5. 4 V
6. 5 V
7. 6 V
8. 9 V
9. None of the above
10. Impossible to determine



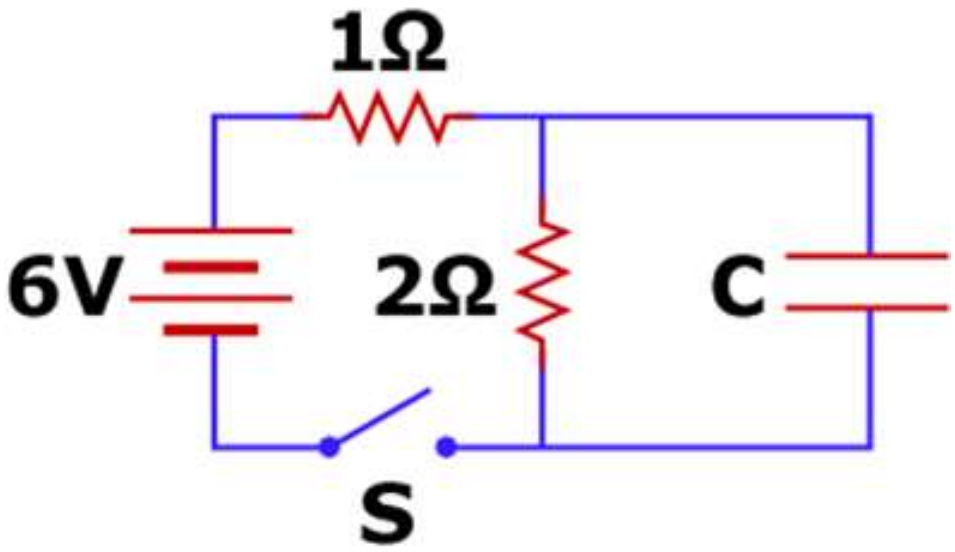
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Assessing to Learn (19 of 21)

Consider the circuit below. Switch S is closed at $t = 0$. What is the current through capacitor C just after the switch is closed?

1. 0
2. 1 A
3. 2 A
4. 3 A
5. 4 A
6. 5 A
7. 6 A
8. 9 A
9. None of the above
10. Impossible to determine



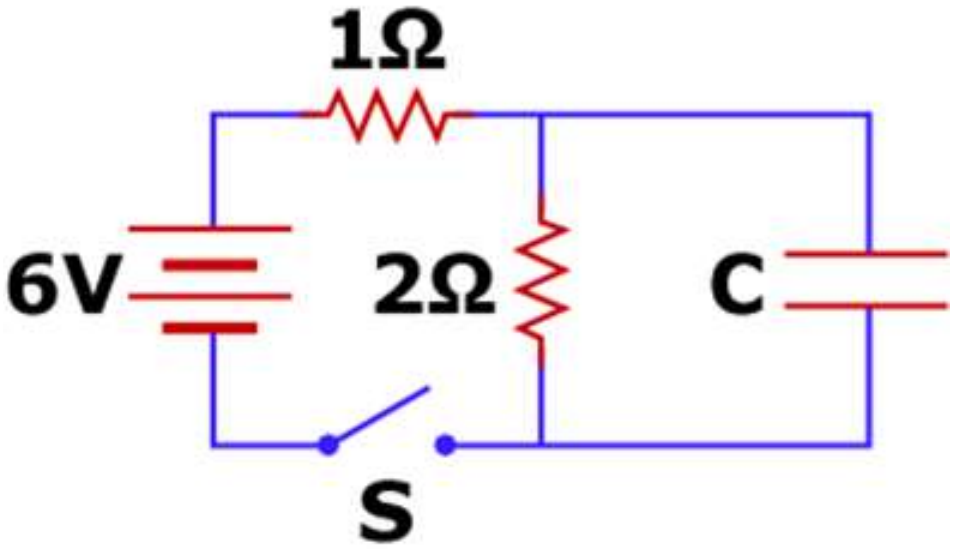
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Assessing to Learn (20 of 21)

Consider the circuit below. Switch S is closed at $t = 0$. What is the voltage across the capacitor C long after the switch is closed?

1. 0
2. 1 V
3. 2 V
4. 3 V
5. 4 V
6. 5 V
7. 6 V
8. 9 V
9. None of the above
10. Impossible to determine



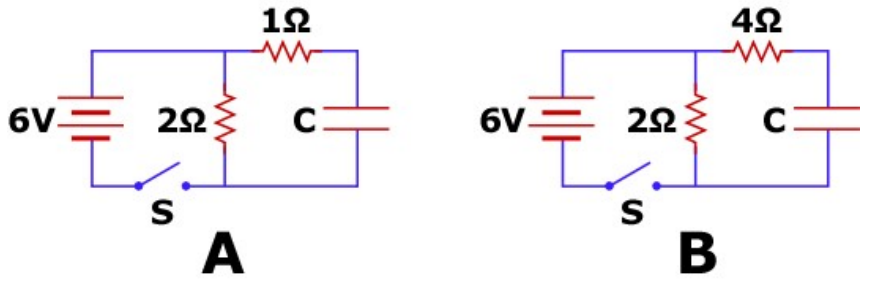
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Assessing to Learn (21 of 21)

Consider the following circuits. The batteries and capacitors are identical, and the switches have been closed for a very long time. At $t = 0$, both switches are opened. Which pair of resistors dissipates more total energy?

1. The resistors in A dissipate more energy.
2. The resistors in B dissipate more energy.
3. Both pairs of resistors dissipate the same amount of energy.
4. The amounts of energy dissipated cannot be compared.



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